

Can technology free time? Experimental evidence from an AI chatbot helping caseworkers solve employers' hiring difficulties

Jeremias KLAEUI

Joseph BARREAU

Manon BOUJU

Roland RATHELOT *

January 2026, *Preliminary version, please do not quote*

Abstract

In late 2024, the French public employment service introduced a new AI tool to support caseworkers dedicated to assisting client recruiters in their hiring decisions. Caseworkers support employers through two channels: screening applications from job seekers who apply on their own, and sourcing candidates who have not applied. The tool was designed to help caseworkers when sourcing candidates, and save them time for other tasks. We leverage the randomised implementation of this intervention to measure its impact on job-filling probabilities. We find that the intervention successfully increased the probability of filling posted jobs by between 10 and 15 per cent. While the intervention was intended solely to facilitate the sourcing of job seekers who had not initially applied for the job, we find that the increase in hires comes from both job seekers who were sourced and job seekers who had applied on their own and were screened by the caseworker. Our interpretation is that the AI tool freed up caseworkers' time, allowing them to devote more time to screening applications (and other tasks).

Keywords: job search, hiring, public employment service, artificial intelligence, workers' performance

JEL Codes: J2, J6, J7

*Klaeui and Rathelot: Institut Polytechnique de Paris. Barreau and Bouju: France Travail. We would like to thank Guillaume Bied, Bruno Crépon, Cyril Nouveau, and Michèle Sebag for helpful comments. We thank France Travail for entering in this research partnership and providing access to data and computing facilities. We thank Louise Demoury for superb research assistance. Funded by the European Union (ERC INASHI grant agreement 101087581). Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Research Council. Neither the European Union nor the granting authority can be held responsible for them. All errors are our own. Klaeui is also affiliated to CREST and KOF Swiss Economic Institute (ETH Zurich), Rathelot to CREST, CEPR, CESifo, IZA, J-PAL, RFBerlin.

1 Introduction

Most countries experience both persistent and sizeable levels of unemployment and unfilled vacancies. The co-existence of these two phenomena can be explained either by the existence of mismatch, whether geographic or skill-related, or by frictions in the matching process. A large strand of the literature has explored and broadly rejected the idea that mismatch could be a sizeable driver of persistent unemployment and hiring difficulties, leaving frictions as the most likely determinant. Frictions themselves can be very diverse. Some frictions are rooted in a lack of information on the part of job seekers about existing vacancies, or about whether they are a good fit for these vacancies. Public Employment Services deploy many services (e.g. job-search assistance) to guide job seekers in their search. Policies as well as interventions aiming to provide more information to caseworkers have been evaluated by a large number of papers, and are shown to improve re-employment outcomes at the micro level (Card et al., 2018; Altmann et al., 2018; Schiprowski, 2020; Belot et al., 2022; Altmann et al., 2023; Le Barbanchon et al., 2023; Behaghel et al., 2024; Cederlöf et al., 2025), but questions remain as to whether they are effective in solving imbalances between labour demand and labour supply at the macro level (Crépon et al., 2013; Gautier et al., 2018; Cheung et al., 2025).

In this paper, we measure the impact of an intervention that aims to support caseworkers in providing services to recruiters during the hiring process. We believe this kind of intervention is relevant for several reasons. First, because the demand side is more and more often on the short side of labour markets, alleviating frictions stemming from information imperfections faced by employers may be more promising in reducing labour market imbalances. Second, the literature provides much less evidence on the impact of interventions aiming to improve the information set of employers than on interventions targeting job seekers (with the notable exceptions of Pallais (2014); Algan et al. (2022) in developed countries and Carranza et al. (2022); Abel et al. (2020) in developing countries). Third, the existence of duration dependence (e.g. Kroft et al. (2013)) or discrimination in labour markets can be interpreted as indirect evidence that employers find it very costly to screen or source relevant candidates. Fourth, public employment services in several countries are currently ramping up the provision of services to recruiters. It is therefore worth understanding how this support can be provided in the most effective manner.

This paper takes place in France, where the public employment service, France Travail, has already deployed a fairly comprehensive set of services to firms. Some caseworkers are specifically dedicated to supporting employers. Employers posting a job adver-

tisement at France Travail can access a caseworker who helps them to both (i) screen applications that job seekers submit on their own and (ii) source job seekers who have not applied but would be relevant for the job. At the end of the process, after checking both the relevance of candidates and their interest in the job, France Travail caseworkers provide recruiters with a short list of candidates. These activities are time-consuming. The intervention we study in this paper aims to support the work of caseworkers when they perform the sourcing task: instead of having to call or email job seekers to check their availability, their interest, and their fit, caseworkers are provided access to an AI chatbot service, *MatchFT*, which engages in an SMS conversation with up to fifty job seekers selected by caseworkers, answers their queries, and records both their potential interest and their fit. This service is designed to free up caseworkers' time, which they can dedicate to other tasks.

To evaluate the impact of this intervention, France Travail designed a randomised controlled trial. Employers were randomised into two groups: (i) the treatment group, which benefits from the *MatchFT* chatbot service, and (ii) the control group, which does not and remains under business as usual. Caseworkers are free to use the service or not. Take-up was far from systematic: we observe that *MatchFT* was used for only between 30% and 40% of eligible jobs. Our main outcome of interest is the probability that a posted job is filled by a job seeker registered at France Travail. Treated firms hired 10% to 15% more job seekers via France Travail than control ones, an impact that is quantitatively large considering the take-up rate.

Our data allow us to know whether hired workers were screened (i.e. they applied to the job on their own and were shortlisted by caseworkers) or sourced (i.e. they had not applied before caseworkers contacted them about the job). While the intervention only reduces the cost of sourcing, we find that the increase in the number of hires comes half from sourced workers and half from screened workers. One way to interpret this result is that *MatchFT* freed up time that caseworkers used for other tasks, in particular screening candidates for the same job advertisement.

We explore other sources of spillovers by leveraging the fact that, treatment being randomly assigned at the employer level, the share of job advertisements eligible to *MatchFT* that a caseworker had to handle was as good as random. Interestingly, we find that an increase in the share of treated job advertisements is correlated with an increase in the number of hires experienced by control job advertisements. This means that control job advertisements benefitted from being handled by caseworkers who had more time freed up by the intervention. This interpretation is supported by a caseworker survey, in which

time savings emerged as the most frequently cited benefit of the tool—reported by 56% of users, ahead of improvements in service quality. Another consequence of this result is that the SUTVA (Stable Unit Treatment Value Assumption) is unlikely to hold, and that the treatment effect we measure when comparing treated and control job advertisements is likely to be a lower bound of the true impact.

Our paper contributes to four strands of the literature. First, the literature that aims to measure and understand the consequences of information frictions faced by firms when hiring (Pallais, 2014; Abel et al., 2020; Carranza et al., 2022; Algan et al., 2022). Our first contribution is to reassert the importance of information frictions faced by firms: tools that improve the performance of caseworkers in charge in supporting employers have the potential to improve hiring outcomes.

Second, the literature that aims to measure the importance of caseworkers in the matching process between labour supply and labour demand (Schiprowski, 2020; Cederlöf et al., 2025). We contribute by showing that services provided by caseworkers can be enhanced and effectively complemented by AI tools.

Third, the literature on how artificial intelligence can be leveraged to help match supply and demand in labour markets (and other markets) (Bied et al., 2023; Le Barbanchon et al., 2023; Behaghel et al., 2024). While most papers in this literature evaluate the impact of AI-based recommender systems that provide recommendations to job seekers, this paper shows that AI tools can also be used to save human time and to substitute for routine tasks (such as scheduling meetings or extracting basic information).

Fourth, the literature about the performance of HR officers and civil servants. A strand of literature has looked into the influence of selection devices, at the influence of managers or at incentives to increase performance among civil servants (Dal Bó et al., 2013; Biasi, 2021; Muñoz and Prem, 2024; Muñoz and Otero, 2025). Hoffman et al. (2018) shows that the introduction of a guiding tool for HR officers improves their decisions. Our contribution is to show that it is possible to use AI tools to effectively complement workers' actions, freeing up their time on some tasks to increase their overall performance.

The remainder is organised as follows. We introduce the setting and the experimental design in the next section. Section 3 presents the data and descriptive statistics. In Section 4, we explain the methodology and provide the main results on the impact of the intervention on hiring outcomes. In Section 5, we explore the mechanisms.

2 Setting

2.1 Institutional setting

France Travail is the French public employment service. Its main mission is to help registered job seekers find a job and to help employers match with unemployed job seekers. France Travail maintains a job portal, francetravail.fr, which is the largest job board in the country, with typically around 1 million posted jobs on a given day.

As is the case in many settings, the most common way in which matching takes place is asymmetric: employers post jobs on the job board, job seekers browse the platform and apply to jobs (applications are initiated within the platform but typically finalised on firms' own websites), and employers then choose whom to hire among the applications they receive.

France Travail has a dedicated team of caseworkers specialised in supporting employers. Firms can, upon request, receive support from caseworkers throughout the hiring process, as described in [Algan et al. \(2022\)](#). Most importantly, once they have posted a job advertisement, employers can request assistance from France Travail to help them find suitable candidates for the job; we refer to this support track as “preselection” throughout the text. In this case, caseworkers perform two types of actions:

1. *Screening*. Job seekers can apply to this job advertisement on their own, but the quality of applications may be heterogeneous. One service provided by caseworkers is to screen the pool of applications and shortlist those that are a good fit for the job. Only the shortlisted candidates will be put forward to the firm.
2. *Sourcing*. In addition to screening, caseworkers may and usually do decide to extend the list by sourcing candidates from the profile bank (i.e., all job seekers who have agreed to be part of this bank). Typically, caseworkers first use a matching tool that identifies job seekers in the bank who correspond to the main characteristics of the job advertisement (e.g., occupation, location, experience). They then contact, by email or by phone, the candidates whom they consider to be a good fit.

Caseworkers then compile the list of job seekers they have screened and sourced and hand it over to the firm's recruiter. A survey of caseworkers conducted by France Travail as part of this study confirms the importance of both tasks.¹ Screening objectives

¹The survey was administered in February 2025 to caseworkers in agencies where the experiment was deployed. Out of 86 respondents, 77 answered questions about preselection objectives. Full results are reported in Appendix B.1.

are nearly universal: 100% of caseworkers cite avoiding irrelevant applications reaching employers as an objective, and 94% mention filtering candidates on attractive vacancies. Sourcing is also widespread, with 88% citing the search for candidates on hard-to-fill vacancies as an objective.

A companion survey conducted in non-experimental regions provides further detail on caseworker practices.² Upon receiving a vacancy in preselection mode, 80% of caseworkers begin searching for candidates immediately rather than waiting for applications to arrive. To identify suitable profiles, caseworkers rely primarily on CVs (98% consult them always or often) and job seeker dossiers (92%). Most caseworkers (78%) report targeting candidates narrowly rather than contacting broadly, with 90% typically reaching out to 20 candidates or fewer per vacancy. The time costs of the sourcing process are substantial. Caseworkers spend 1–5 minutes initially reviewing each candidate, yet only follow up with roughly 3 out of every 10 who respond. Furthermore, the primary reason for the follow-up call is to verify availability (95%), followed by obtaining additional profile information (94%). This suggests considerable effort is spent on candidates who may ultimately be unavailable — a friction the MatchFT tool is designed to address.

2.2 Intervention

Late 2024, France Travail started to roll out *MatchFT*, a new tool made available to caseworkers to help them source job seekers for jobs in preselection. This tool consists in an AI-driven SMS chatbot that is designed to automatically probe the interest and the availability of a job seeker for a given job, assess how well they would fit the job, and answer questions that job seekers may have about the contents of the job.

In practice, when a caseworker uses the matching tool to get a list of potential suitable candidates in the bank of profiles, the tool provides a button in front of the name of each candidate. When the caseworker presses a button, an SMS conversation between the chatbot and the candidate starts. The tool then summarises the content of the conversation (e.g., whether the candidate is interested or not, whether they are skilled for the job or not) within the usual dashboard of caseworkers.

The main objective of the tool is to save caseworkers' time, by substituting potentially time-consuming email exchanges or phone conversations.

²This survey was administered to caseworkers in agencies where MatchFT was not deployed. N=91 respondents.

2.3 Experimental protocol

The random assignment into treatment and control was conducted by France Travail using a stratified sampling procedure. The sampling frame consisted of all registered establishments that had published at least one job offer since 2019. The primary stratification variable was the number of job offers published in 2023. France Travail first estimated the expected number of job postings in 2023 using regression models based on establishment-level characteristics. For establishments with complete information, the predicted value was obtained from a linear regression estimating the probability of being active (i.e., publishing at least one ad in 2023), using the following covariates: establishment age, sector of activity, number of job offers published in 2022, number of job offers published between July and December 2022, region, and employment size category. For establishments with missing values, a feature of the employment size and region variables, France Travail ran complementary regressions excluding the missing variables.

For each regression – full information sample, missing establishment size, missing region – establishments were grouped into deciles based on the predicted number of postings from their respective regression. The random assignment was carried out within each decile stratum. Establishments with missing values were not excluded but were stratified separately, according to the deciles derived from the corresponding complementary regression.

3 Data

3.1 Data sources

3.1.1 France Travail administrative data

We rely on job advertisements posted at France Travail. For each job advertisement, we observe the occupation, the place of work, the employer, the date of publication, and the date on which the advertisement is closed. Every time a job ad is posted at France Travail, a proprietary algorithm classifies it as either easy or hard to fill, based on its characteristics (e.g., occupation, location). We use this variable (which we sometimes refer to as *IEE score*) to study the heterogeneity of the treatment effect with respect to the extent of hiring difficulty.

This paper focuses on job advertisements that employers have chosen to place in preselection (one of the hiring assistance services provided by France Travail). In the adminis-

trative data, we observe the date when the job advertisement is placed in preselection, as well as the identifier of the caseworker (and the agency) handling the job advertisement.

For jobs in preselection, we observe which job seekers have been shortlisted by caseworkers (and hence delivered to recruiters). We also observe applications made by job seekers on their own, whether they have been shortlisted or screened out by caseworkers. For sourcing in the control group (business as usual), we do not observe the universe of job seekers who have been sourced if they are not shortlisted. For sourcing in the treatment group, we observe when job seekers have been contacted via the *MatchFT* chatbot, even when they are not shortlisted subsequently.

3.1.2 Déclaration Sociale Nominative (DSN)

We use administrative employer–employee data drawn from the *Déclaration Sociale Nominative* (DSN), the French mandatory monthly social declaration system. The DSN requires all private-sector employers to report, on a continuous basis, detailed information on their employees and payroll to the social security administration. DSN declarations are collected and processed by French social security institutions.

The DSN provides exhaustive, individual-level information on employment relationships, including gross and net earnings, hours worked, employment contracts, occupational status, firm identifiers, and key demographic characteristics of workers. Data are reported at a monthly frequency and cover virtually the entire population of private-sector employees in France.

The DSN is transmitted to France Travail and matched, at the individual and employer (establishment) level, with the job advertisement data and the application data. We use it to detect matches between a job seeker and an employer following an application.

3.2 Outcomes

3.2.1 Job filling

The main outcome of interest is the extent to which a posted job in preselection is filled, and in particular, filled by a registered job seeker. By definition, jobs in the preselection track can only be filled (within the France Travail platform) by applicants who are shortlisted by caseworkers, as employers do not observe other potential France Travail applicants. We construct our main outcome at the job-advertisement level by linking

caseworker shortlists to hires using administrative data from the DSN. For each job advertisement, we identify hires of candidates who had been shortlisted by caseworkers. We match DSN work contracts to the preselection shortlist using establishment and individual identifiers. We restrict attention to hires with contract start dates on or after the date of shortlist creation. For each advertisement, we then count the number of hires that begin within a given time window (e.g., 15 or 30 days) after the start of the preselection process. This outcome includes all hires corresponding to shortlisted candidates, regardless of whether they came from the *MatchFT* tool.

3.2.2 Shortlisted candidates

To shed light on the mechanisms behind the impact of the intervention, we use data on shortlisted candidates. First, we study the impact on the number of candidates shortlisted. Second, we investigate whether shortlisted candidates came from the screening process (i.e., they applied to the job on their own and were selected by caseworkers) or the sourcing process (i.e., they had applied to the job but were contacted by caseworkers and nudged into applying).

3.3 Descriptive statistics

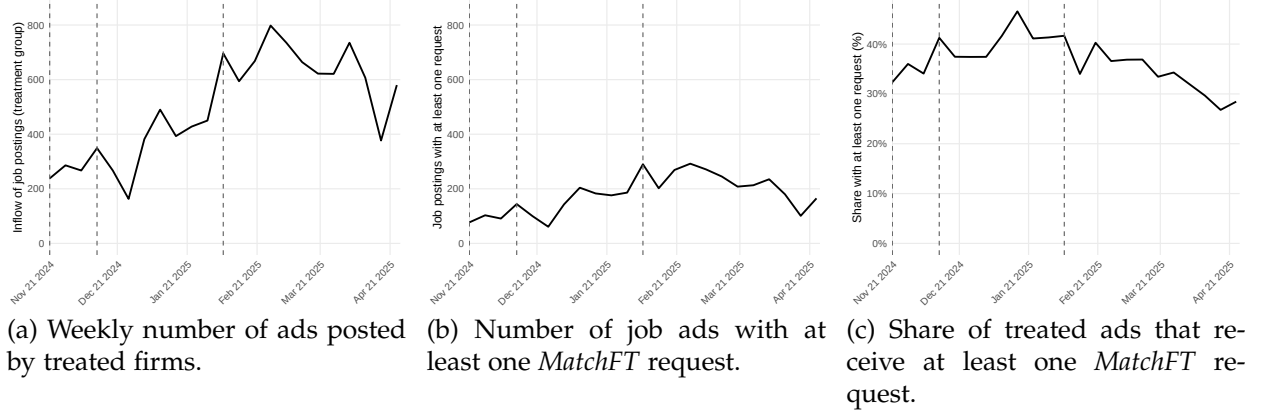
3.3.1 Experiment roll-out and take-up

Figure 1 shows the evolution of take-up of the *MatchFT* tool. Panel (a) shows the total weekly number of job advertisements in preselection that are assigned to the treatment, i.e. the eligible population. This number increases from around 300 in late 2024 to around 700 in February–March 2025, largely reflecting the increase in the number of agencies involved in the experiment. Panel (b) shows the weekly number of job advertisements with at least one *MatchFT* request, which lies between 100 and 300. Panel (c) shows the ratio between the two panels: the share of eligible advertisements for which we observe at least one *MatchFT* request. This ratio is fairly stable at around 40%, before steadily decreasing to 30% by the end of the period.

3.3.2 Sample description and balancing checks

Table 1 presents summary statistics for the control group in terms of baseline characteristics. All variables are measured over the 12-week period preceding treatment assignment.

Figure 1: Evolution of take-up of the *MatchFT* service among treated job ads.



Notes: The date refers to the day at which the preselection process started for the ad. Vertical dashed lines mark the roll-out dates: November 21, 2024 (first roll-out in a subset of agencies), December 12, 2024 (roll-out in the remainder of agencies), and February 6, 2025 (increase of treatment share to 75%).

To check the validity of the randomisation process, we also show baseline differences between treated and control establishments. Differences between treated and control groups are estimated using the regression specification in Equation (1).

$$Y_{ies} = \alpha + \beta \cdot \text{Treated}_i + \delta_e + \lambda_s + \varepsilon_{ies}, \quad (1)$$

where Y_{ies} denotes the cumulative outcome for establishment i in subexperiment e and strata s ; Treated_i is an indicator for whether the establishment was treated in that subexperiment; δ_e and λ_s are subexperiment and strata fixed effects, respectively; and ε_{ies} is the error term.

3.3.3 Usage of the *MatchFT* tool

In Table 2, we investigate how caseworkers used the *MatchFT* tool when they used it. In total, 110,137 chatbot interactions were initiated across 4,151 job advertisements. Conditional on sending at least one *MatchFT* request for a given job advertisement, caseworkers used the tool on 27 job seekers on average.

Caseworkers can choose which criteria they want the chatbot to assess. Interest in the job is compulsory (present in 100% of requests), but availability (83%), adequacy with tasks (61%), possible mobility (61%), and relevant experience (57%) were also frequently assessed by caseworkers.

Did job seekers engage with the chatbot? Ultimately, only 5% of requests concerning

Table 1: Balance check and descriptive statistics (pre-experiment period)

	Always control (N = 2141) Mean	Always treated (N = 4308) Diff (p)	Second roll-out (N = 2158) Diff (p)
Establishment size			
10 or less	0.449	-0.000 (0.976)	0.003 (0.822)
11–25	0.138	0.003 (0.753)	0.012 (0.267)
26–50	0.156	-0.009 (0.366)	0.009 (0.423)
More than 50	0.183	-0.001 (0.950)	-0.023** (0.048)
Missing	0.075	0.007 (0.340)	-0.002 (0.850)
Establishment industry			
Construction	0.069	0.001 (0.857)	-0.010 (0.168)
Manufacturing	0.116	0.006 (0.464)	0.001 (0.924)
Commerce	0.156	-0.005 (0.575)	0.004 (0.728)
Services	0.150	0.005 (0.606)	0.010 (0.369)
Health	0.140	-0.004 (0.654)	-0.002 (0.848)
Ad posting & caseworker referrals			
Ads posted	13.470	-0.216 (0.896)	-1.697 (0.308)
Ads posted (pre-sel.)	2.259	-0.113 (0.458)	-0.127 (0.429)
Candidates shortlisted	17.028	-1.311 (0.236)	-1.335 (0.265)
Hiring			
Hires via platform	0.708	0.010 (0.882)	-0.032 (0.657)

interest resulted in a positive response (i.e., job seekers did not ignore the tool and responded positively). This implies that the tool generated, on average, 1.3 interested candidates per job advertisement when it was used.

Caseworkers appear to have taken the information provided by *MatchFT* seriously. Forty-three per cent of candidates who indicated interest in the job via the chatbot were ultimately shortlisted by caseworkers, compared with 2% of candidates who responded negatively or did not respond (and may have been contacted independently by caseworkers).

Table 2: Usage of *MatchFT*

Criteria	Asked in N ads	Share	N requests	Positive answers	Referrals among pos. answers	Referrals among neg. answers
Interest	4132	100.0%	109415	5.0%	42.7%	1.5%
Availability	3448	83.4%	87746	4.1%	50.4%	1.8%
Tasks	2617	63.3%	67781	3.7%	54.1%	1.9%
Mobility	2518	60.9%	67736	3.6%	50.6%	1.7%
Experience	2369	57.3%	63615	3.3%	52.3%	1.9%
Working hours	1590	38.5%	35280	3.6%	49.5%	2.1%
Wage expectations	1415	34.2%	33559	3.6%	48.0%	2.0%

Notes: Summary statistics for job postings in treatment group with at least one *MatchFT* request. No answers ('Answer pending') are counted as negative answers.

3.4 Screening vs. sourcing

Table 3 compares the number of shortlisted candidates per posting for all postings with those for which we observe at least one *MatchFT* request. This comparison cannot be interpreted causally, as the two sets of postings have no reason to be similar.

In the sample of all preselection postings, we observe that caseworkers shortlist an average of eight candidates, roughly six coming from screening and two from sourcing. Postings with at least one *MatchFT* request have a slightly longer shortlist (around ten candidates) and a higher share of shortlisted candidates coming from sourcing.

Table 3: Number of shortlisted candidates per posting

	All postings		Postings with MatchFT req.	
	Mean	Median	Mean	Median
Total candidates per posting	7.7	5.0	7.6	5.0
Via sourcing	2.2	1.0	2.2	2.0
- out of which after MatchFT request	0.7	0.0	1.2	1.0
Via screening	5.6	3.0	5.3	3.0

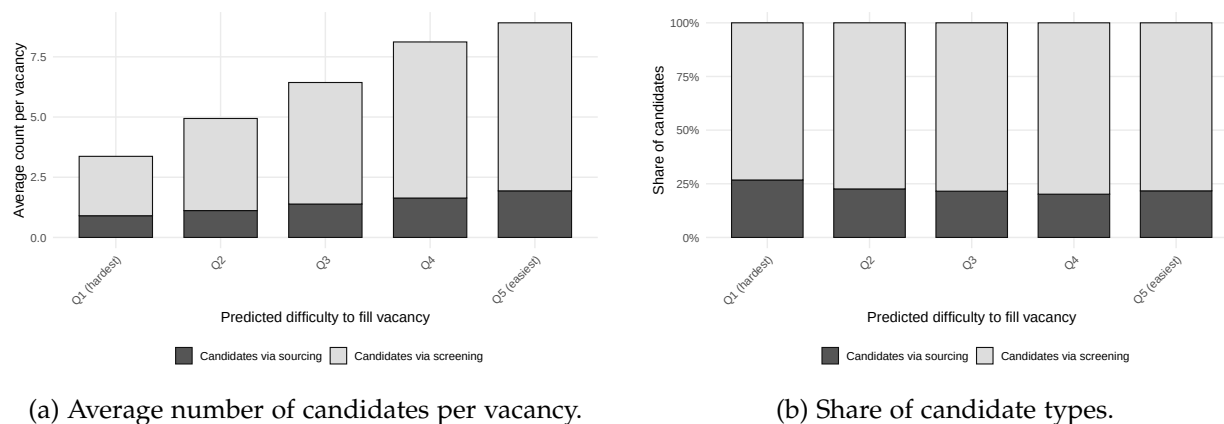
Notes: Summary statistics for job postings in treatment group with at least one referral

Figure 2 shows how the number and composition of shortlisted candidates vary across quintiles of the IEE score, which rates how easy a job is to fill. First, we assess the relevance of this score. Jobs in the 20% hardest to fill receive, on average, three shortlisted candidates, while those in the 20% easiest to fill receive an average of nine.

While the total number of shortlisted candidates varies, the share coming from sourcing versus screening does not, and remains between 20% and 25% across all quintiles. Is this result counter-intuitive? One might have expected that, for vacancies that are easier to fill, there are enough job seekers who apply on their own, so that caseworkers do not need to engage in sourcing. Conversely, for jobs that are hard to fill, there are fewer applicants, and caseworkers may want to compensate by performing more sourcing. This does not appear to be the case, possibly because the returns to sourcing also decrease with the difficulty of filling the vacancy. Furthermore, it is consistent with the survey evidence cited in Section 2.1: Upon starting the recruitment process for a vacancy in pre-selection mode, 80% of caseworkers begin searching for candidates immediately rather than waiting for applications to arrive.

Figure 3 shows how the number and composition of hired unemployed job seekers vary with how difficult a job is to fill. Again, the number of hires increases with how easy

Figure 2: Composition of shortlisted candidates by predicted vacancy filling difficulty (control group).

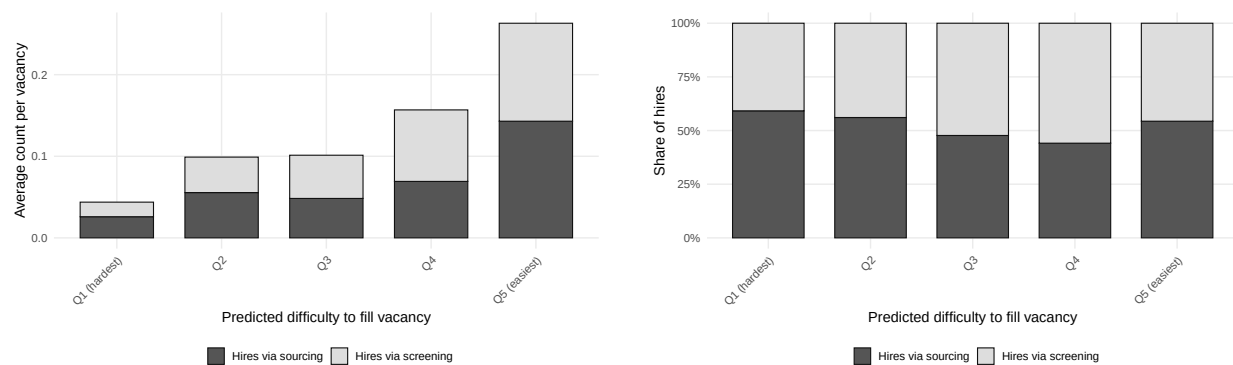


Notes: Sample restricted to control group vacancies with non-missing IEE score. Quintiles are based on the IEE score, a prediction of the probability of filling the vacancy within 30 days computed by France Travail, where Q1 contains the hardest-to-fill vacancies and Q5 the easiest. Candidates via sourcing are caseworker-initiated contacts or applications following a *MatchFT* recommendation. Candidates via screening are job seeker applications approved as referrals by the caseworker, excluding those triggered by *MatchFT* recommendations. Counts are measured within 30 days of the vacancy entering the preselection process.

the job is to fill, which is intuitive. In addition, the share of hired job seekers who were screened versus sourced is constant across quintiles. Interestingly, the share of sourced job seekers among those hired is around 50%, while sourced shortlisted applicants represent only about 25%. This suggests that, all else equal, sourced workers are likely to be of higher quality than screened ones.³

³An alternative explanation, which we cannot exclude, is that this difference comes from workers' behaviour, whereby sourced workers are more likely to accept the employer's offer.

Figure 3: Composition of hires via portal by predicted vacancy filling difficulty (control group).



(a) Average number of hires via platform within 30 days of recruitment start.

(b) Share of hire types.

Notes: Sample restricted to control group vacancies with non-missing IEE score. Quintiles are based on the IEE score, a prediction of the probability of filling the vacancy within 30 days computed by France Travail, where Q1 contains the hardest-to-fill vacancies and Q5 the easiest. Hires via sourcing are hires of candidates referred through caseworker-initiated contacts or *MatchFT* recommendations. Hires via screening are hires of candidates referred through caseworker-approved job seeker applications. Hires are measured within 45 days of the vacancy entering the preselection process.

4 The impact of reducing caseworkers' screening costs on hiring outcomes

4.1 Methodology

Our baseline analysis is conducted at the advertisement level. We consider all ads that entered the preselection stage and define the outcome variable as an indicator for whether a hire was made through the PES within a given number of days after the recruitment process has started. Note that the outcome entails all hires made from the candidate shortlist, no matter whether the candidate was found through sourcing or screening and irrespective of whether the sourcing was conducted via MatchFT or calls or emails.

A key empirical challenge arises from the fact that both ad-level filling rates and the rollout of the treatment vary over time. This raises concerns about potential confounding due to time trends in labour demand or supply.

To address this concerns, we control for baseline differences in fill rates across strata and over time. Specifically, we include fixed effects for each stratum interacted with an indicator for the period following the main increase in treatment exposure, which occurred on February 6, 2025. This strategy allows us to flexibly account for time-varying differences across strata unrelated to the treatment.

Formally, we estimate the following specification:

$$Y_i = \beta D_{f(i)} + \theta_{s(i)} \times \mathbb{1}[t(i) \geq \text{Feb 6, 2025}] + \varepsilon_i,$$

where Y_i is the binary outcome for ad i , $D_{f(i)}$ is a treatment indicator for the firm or establishment associated with ad i , $\theta_{s(i)}$ denotes fixed effects for the stratum s to which ad i belongs, and the interaction term captures post-period variation in baseline outcomes across strata. Standard errors are clustered at the establishment level.

When estimating treatment effects, we account for the stratified random assignment by including strata fixed effects. However, conventional standard errors may be conservative in the context of stratified random assignment when the treatment probability within strata deviates from 0.5 (Bugni et al., 2018). To address this, we apply the correction proposed by Bugni et al. (2018) using the `sreg` package for R (Trifonov et al., 2024). To account for the fact that treatment was assigned at the establishment level and some establishments switch treatment status on February 6, we aggregate outcomes

at the establishment-by-subexperiment level. Specifically, we compute the average outcome for each establishment across all job postings, at each evaluation horizon, within each subexperiment. This avoids the need for clustered standard errors and ensures correct alignment between the unit of treatment assignment and the unit of observation in the estimation. In the mechanism section, when outcomes are count data (such as the number of shortlisted candidates), we will apply Pseudo Poisson Maximum Likelihood regressions instead of the linear regressions with corrected standard errors. To increase power, we will control additionally for unemployment office $\times$ subexperiment fixed effects.

4.2 Impact on job-filling probability

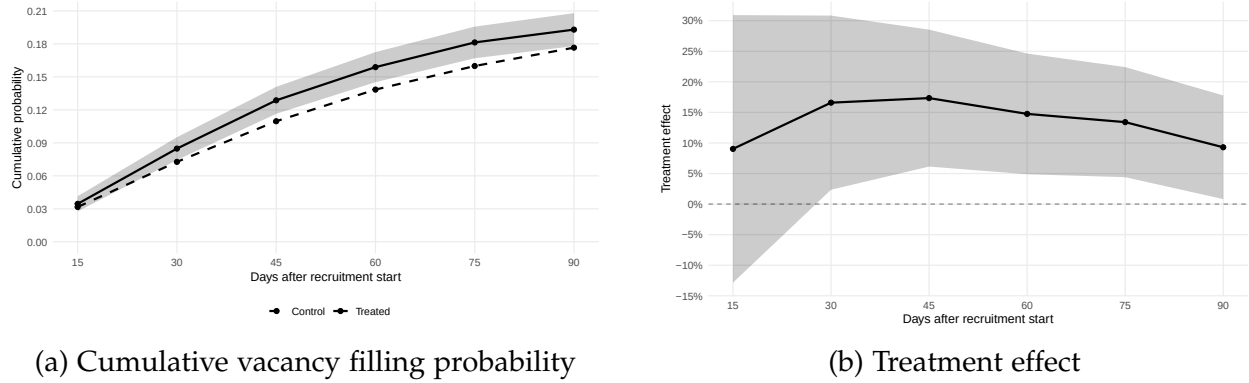
Figure 4 presents our main results on the treatment effect of the matching tool on vacancy filling through the PES. Panel (a) displays the cumulative probability of hiring a candidate within a given number of days after the vacancy enters preselection mode. By day 45, treated vacancies have a hiring probability of approximately 13%, compared to 10% for control vacancies.

Panel (b) expresses these effects as a percentage of the control mean. The treatment effect peaks at approximately 15% around day 45, after which control vacancies begin to catch up. This pattern suggests that the matching tool generates both speed and level effects: the larger relative effect at earlier horizons reflects faster hiring for treated vacancies, while the persistent and statistically significant effect at 90 days indicates that some of these additional hires might not have occurred absent treatment. Table 4 reports the corresponding estimates.

Our results are robust to alternative specifications. Tables A1–A3 in the Appendix show that the estimates are stable when using only strata fixed effects or when including additional controls for the number of positions and pre-experiment hiring history. Table A4 presents Pseudo-Poisson-Maximum-Likelihood (PPML) estimates for the number of hires, which yield similar conclusions. Table A5 shows that results also hold when using an alternative outcome measuring whether all advertised positions were filled.

To validate our experimental design, we conduct a placebo test using vacancies from November 2023 to April 2024, a year before the experiment began. We assign placebo treatment status based on establishments’ eventual treatment assignment. Figure A2 and Table A8 in the Appendix show no significant differences between treatment and control groups in the pre-treatment period, confirming the absence of differential pre-trends.

Figure 4: Treatment Effect on Vacancy Filling



Notes: Panel (a) shows the cumulative probability of filling a vacancy through the PES within the specified number of days after recruitment start, separately for treated and control vacancies. Panel (b) shows the treatment effect as a percentage of the control mean at each horizon. Shaded areas and error bars represent 95% confidence intervals.

Table 4: Treatment Effect on Vacancy Filling

Horizon:	Vacancy filled					
	15 days	30 days	45 days	60 days	75 days	90 days
Treated	0.0029 (0.0035)	0.0121** (0.0053)	0.0190*** (0.0063)	0.0204*** (0.0070)	0.0214*** (0.0073)	0.0164** (0.0076)
Strata $\times$ sub-experiment FE	X	X	X	X	X	X
Observations	10,166	10,166	10,166	10,166	10,166	10,166
Control mean	0.0317	0.0727	0.1097	0.1384	0.1599	0.1766

Notes: Results from horizon-specific linear regressions. Outcome: binary indicator of hire through the PES for the job posting. One observation is one establishment, if an establishment has posted more than one posting, the filling rates are averaged at the horizon level. Subexperiment denotes an indicator variable whether the preselection process for the vacancy started after Feb 6, 2025 when the treatment share was increased from 50 to 75%. The outcome is averaged at the establishment $\times$ subexperiment level and the standard errors are adjusted following [Bugni et al. \(2018\)](#).

5 Mechanisms

5.1 Number of candidates shortlisted

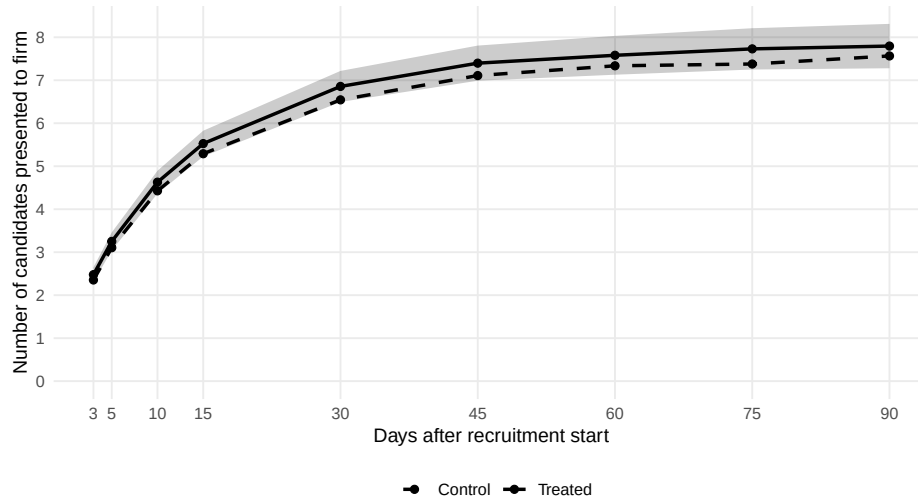
A natural first channel through which the matching tool could improve hiring outcomes is by increasing the number of candidates presented to employers. If caseworkers use the tool’s recommendations to identify additional suitable candidates, one might expect treated vacancies to receive more referrals. Figure 5 shows that the matching tool has a minimal, statistically insignificant effect on the number of candidates presented to firms. Point estimates are small and indistinguishable from zero across all time horizons at the 5% level. By day 45, both treated and control vacancies receive approximately 7.1 candidates on average, with treatment effects of less than 5%. This null result on quantity is consistent with an institutional feature of France Travail’s service model. Caseworkers typically set themselves a target of 5–10 candidates to present per vacancy, aiming to provide employers with a manageable shortlist rather than overwhelming them with applications (Algan et al., 2022).

The absence of a quantity effect, combined with the significant improvement in hiring outcomes documented in Figure 4, suggests that the matching tool operates primarily through a quality channel: caseworkers present candidates who are better matched to vacancies, rather than simply presenting more candidates. Table A9 in the Appendix reports the full regression results. In the following subsections, we investigate how this quality improvement manifests across two distinct referral channels—caseworker-initiated sourcing versus screening of job seeker applications—and how the tool’s effectiveness varies with the caseworkers workload and the vacancy’s predicted filling difficulty.

5.2 Sourcing vs. screening

The matching tool could improve hiring outcomes through two distinct channels. Figure 6 decomposes the treatment effect by those two channels. Panels (a) and (b) focus on hires via sourcing, defined as caseworker-initiated referrals. The treatment effect on sourcing is positive and statistically significant at the 60 days horizon, with treated vacancies approximately 15% more likely to result in a hire through this channel. Panels (c) and (d) examine hires via screening, where the caseworker approves a job seeker’s application. The treatment effect on screening is highest after 30 days and statistically significantly different from zero from day 30 throughout the whole analysis horizon. Table 5 reports the corresponding estimates.

Figure 5: Treatment Effect on Number of Candidates Shortlisted per Vacancy Posting



Notes: The figure shows the cumulative number of candidates presented to firms through the PES within the specified number of days after recruitment start, separately for treated and control vacancies. Estimates from PPML regressions with strata $\times$ subexperiment and agency $\times$ subexperiment fixed effects. Shaded areas represent 95% confidence intervals computed using the delta method, clustered at the establishment level.

Table 5: Treatment effect by channel (sourcing vs. screening)

Panel A: Hires via sourcing

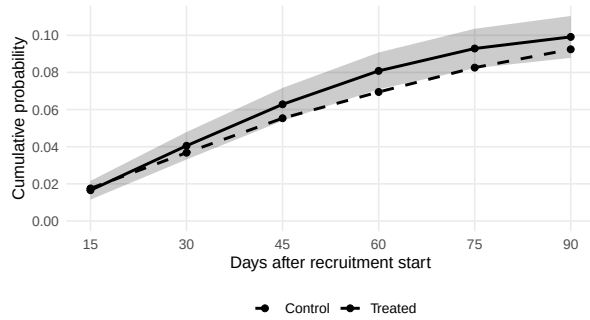
	Vacancy filled					
Horizon:	15 days	30 days	45 days	60 days	75 days	90 days
Treated	-0.0010 (0.0026)	0.0037 (0.0038)	0.0075* (0.0045)	0.0114** (0.0051)	0.0103* (0.0054)	0.0067 (0.0058)
Strata $\times$ sub-experiment FE	X	X	X	X	X	X
Observations	10,166	10,166	10,166	10,166	10,166	10,166
Control mean	0.0175	0.0368	0.0553	0.0695	0.0826	0.0925

Panel B: Hires via screening

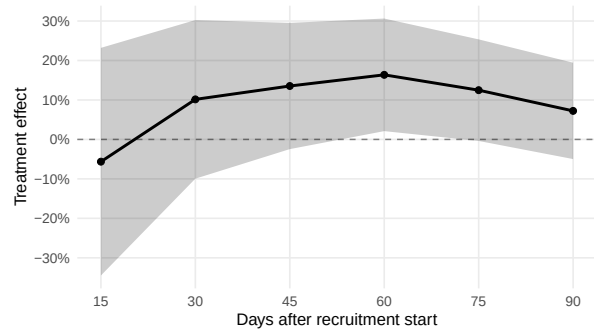
	Vacancy filled					
Horizon:	15 days	30 days	45 days	60 days	75 days	90 days
Treated	0.0037 (0.0025)	0.0094** (0.0039)	0.0125*** (0.0047)	0.0104** (0.0053)	0.0125** (0.0056)	0.0110* (0.0058)
Strata $\times$ sub-experiment FE	X	X	X	X	X	X
Observations	10,166	10,166	10,166	10,166	10,166	10,166
Control mean	0.0146	0.0369	0.0562	0.0725	0.0824	0.0906

Notes: Results from horizon-specific linear regressions. Panel A: outcome is a binary indicator of hire via sourcing (caseworker-initiated referral). Panel B: outcome is a binary indicator of hire via screening (caseworker-approved job seeker application). One observation is one establishment, if an establishment has posted more than one posting, the filling rates are averaged at the horizon level. Subexperiment denotes an indicator variable whether the preselection process for the vacancy started after Feb 6, 2025 when the treatment share was increased from 50 to 75%. The outcome is averaged at the establishment $\times$ subexperiment level and the standard errors are adjusted following Bugni et al. (2018).

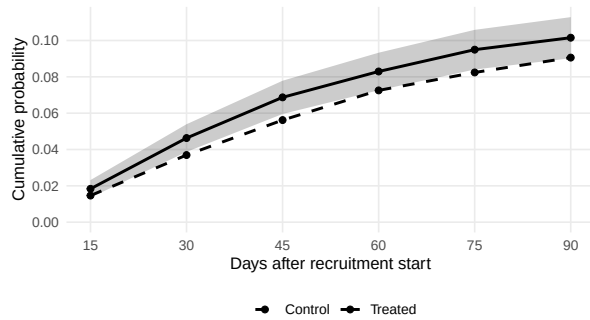
Figure 6: Treatment effect by channel (sourcing vs. screening)



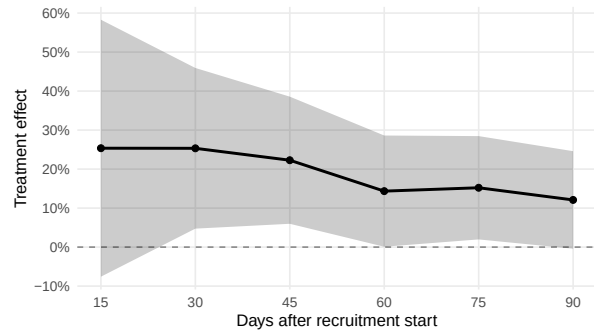
(a) Sourcing – Cumulative probability



(b) Sourcing – Treatment effect



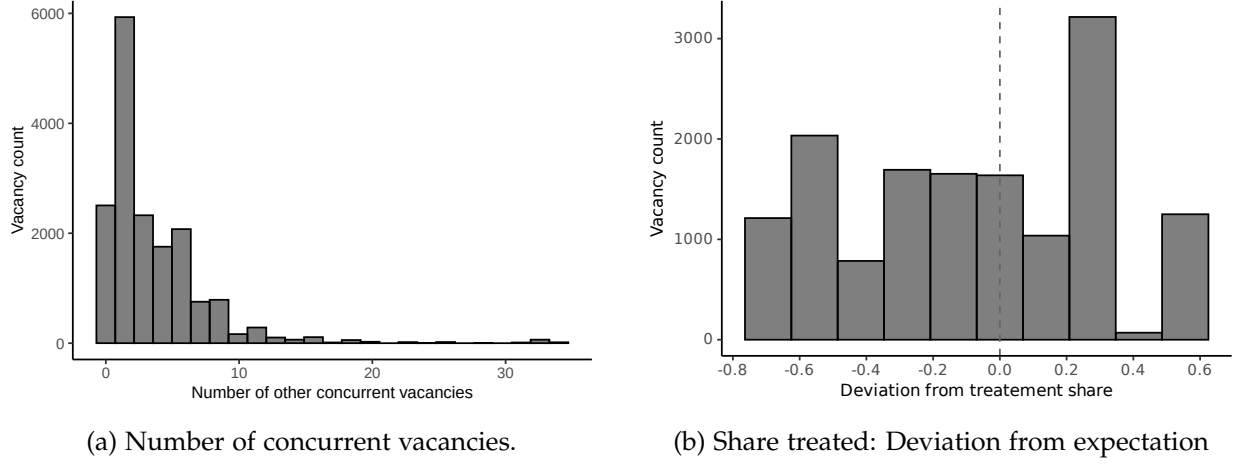
(c) Screening – Cumulative probability



(d) Screening – Treatment effect

Notes: Panels (a) and (b) show treatment effects on hires via sourcing (caseworker-initiated referrals), while panels (c) and (d) show effects on hires via screening (caseworker-approved job seeker applications). Left panels display cumulative hiring probabilities for treated and control vacancies. Right panels display treatment effects as a percentage of the control mean. Shaded areas and error bars represent 95% confidence intervals.

Figure 7: Distribution of caseworker caseload composition.



Notes: Each observation is a vacancy in preselection mode. Panel (a) shows the distribution of the number of other vacancies in preselection mode handled by the same caseworker within a 7-day rolling window centered on the vacancy’s creation date. Panel (b) shows the deviation of the realized treatment share among those concurrent vacancies from the expected share under the experimental design (50% before February 6, 2025; 75% after). The dashed vertical line marks zero deviation.

5.3 Caseworker congestion and time gains

If the matching tool reduces the time caseworkers spend on sourcing for treated vacancies, this time savings may spill over to benefit other vacancies in their caseload. Conversely, the benefits of the matching tool may diminish when they must process many treated vacancies simultaneously. We investigate both possibilities by exploiting variation in the composition of caseworkers’ concurrent caseloads. While the overall treatment share is fixed at 50% (75% after February 6, 2025), the realized share within a caseworker’s concurrent caseload varies due to the small number of vacancies per caseworker per 7-day window.

Figure 7 illustrates the variation underlying our identification strategy. Panel (a) shows the distribution of the number of other vacancies in preselection mode within a caseworker’s 7-day rolling window. The distribution is right-skewed with a median of 2 concurrent vacancies, confirming that caseworkers handle a limited number of vacancies simultaneously. Panel (b) displays the deviation of the realized treatment share among concurrent vacancies from the expected share, that is from 50% if the vacancy was posted during the initial period before February 6 and from 75% if it was posted after February 6 during the increased roll-out period. Despite the fixed aggregate treatment share, the realized share varies substantially at the caseworker level due to the small number

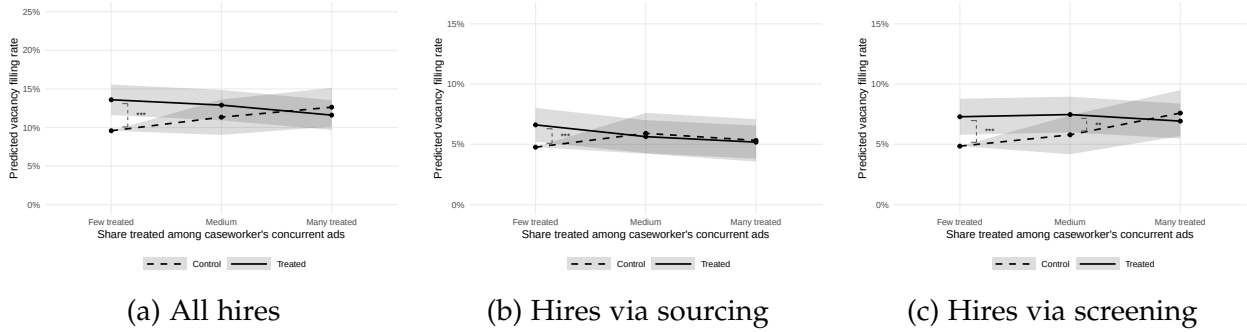
of concurrent vacancies per caseworker, with deviations spanning roughly ± 50 percentage points. This variation is driven by the randomization itself and thus orthogonal to vacancy characteristics conditional on our fixed effects.

We exploit this variation by interacting treatment status with a measure of caseworker-level treatment intensity. For each vacancy, we compute the share of the caseworker's other concurrent vacancies that are assigned to treatment. We then divide this share into terciles, computed separately for the periods before and after the treatment share increase on February 6, and interact these tercile indicators with the vacancy's own treatment status. The omitted category is control vacancies in the lowest tercile of treatment intensity (few treated concurrent vacancies). This yields five coefficients: two for control vacancies exposed to medium and high treatment intensity among their caseworker's other vacancies, and three for treated vacancies at each intensity level of the caseworker's concurrent vacancies.

Table A6 in the Appendix verifies that pre-determined covariates are balanced across the treatment intensity categories. Column (1) shows that the PES' score predicting the likelihood of the vacancy to be filled within 30 days does not differ significantly across any category. Columns (2)–(4) examine the vacancy establishment's hiring history in the year preceding the experiment. The coefficients are generally small and insignificant, with one exception: control vacancies in the medium tercile have 24.4% fewer pre-experiment hires than the reference category ($p < 0.05$), driven by a 34.1% shortfall in the sourcing channel ($p < 0.01$). This imbalance implies that our estimates for this particular cell may understate the true spillover effect, as these establishments had lower hiring activity prior to the experiment. We note, however, that this imbalance does not extend to the high-intensity tercile, nor to any of the treated categories, and is thus unlikely to drive the overall patterns documented below.

Figure 8 displays predicted vacancy filling rates by tercile of treatment intensity among concurrent vacancies, separately for treated (solid line) and control (dashed line) vacancies. Panel (a) shows results for all hires, while Panels (b) and (c) decompose by channel. The results reveal two patterns. First, control vacancies benefit when more of the caseworker's other vacancies are treated: the dashed line slopes upward across terciles from 9.6% to 12.6%, indicating positive spillovers. This is consistent with the interpretation that the matching tool saves caseworker time on treated vacancies, freeing up capacity that benefits control vacancies. Second and slightly less pronounced, the treatment effect—the gap between solid and dashed lines—narrows as treatment intensity increases and is no longer statistically significant in the highest tercile. Panel (b) shows that the treatment effect on sourcing manifests primarily when few of the case-

Figure 8: Within caseworker spillovers: Vacancy filling rates by treatment intensity among concurrent vacancies



Notes: Predicted vacancy filling rates within 45 days based on the OLS estimates reported in Table A7. The x-axis groups vacancies by terciles of the treatment share among the caseworker's concurrent caseload, tercile breaks are computed separately for the periods before and after the treatment share increase on February 6. Concurrent vacancies are defined as other vacancies in preselection mode assigned to the same caseworker within a 7-day rolling window centred on the vacancy's creation date, excluding the vacancy itself. All specifications include strata $\times$ sub-experiment, agency $\times$ sub-experiment, and caseworker fixed effects, and control for the total number of concurrent vacancies and the number in preselection mode. Predictions are computed by adding the estimated coefficients to the mean outcome for control vacancies in the lowest tercile (reference category). Solid lines show treated vacancies; dashed lines show control vacancies. Shaded areas represent 95% confidence intervals. Dashed brackets with significance stars indicate the treatment effect (difference between treated and control) at each tercile. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

worker's other vacancies are treated: the gap between treated and control is significant in the lowest tercile, but as treatment intensity increases, treated vacancies decline slightly while control vacancies rise slightly—neither shift significant on its own—and the treatment effect disappears in the medium and highest terciles. Finally, panel (c) reveals that the screening channel drives the positive spillovers: control vacancies benefit significantly when more of the caseworker's other vacancies are treated. This suggests that time savings from the matching tool on treated vacancies free up caseworker capacity that is reallocated toward screening tasks on other vacancies.

Table A7 in the Appendix reports the underlying regression estimates and adds results at the 90-day horizon. Results are qualitatively similar, albeit less pronounced at the 90-day horizon.

These findings have important implications. The positive spillovers to control vacancies—primarily through the screening channel—suggest that even partial treatment coverage can generate benefits for untreated vacancies as caseworkers reallocate time savings toward screening tasks.

The previous analysis revealed that the matching tool generates positive spillovers to

control vacancies primarily through the screening channel, while the direct treatment effect operates mainly through sourcing. A natural question is whether these patterns vary with vacancy characteristics—in particular, whether the tool’s effectiveness depends on how difficult the vacancy is to fill.

We split the sample at the median of the IEE score, France Travail’s prediction of the probability of filling the vacancy within 30 days. “Easy-to-fill” vacancies are those with above-median predicted fill rates, while “hard-to-fill” vacancies have below-median predictions. We then re-estimate the spillover regressions separately for each group.

Figure 9 and Table 6 present the results. Several patterns emerge. First, at low treatment intensity—when few of the caseworker’s concurrent vacancies are treated—the matching tool generates significant treatment effects across both vacancy types. However, the underlying mechanisms differ.

For easy-to-fill vacancies (left column), the treatment effect operates primarily through the screening channel. Panel (e) shows that control vacancies benefit substantially when more of the caseworker’s other vacancies are treated, with the dashed line sloping upward. This is consistent with time savings from the matching tool being reallocated toward screening tasks. Easy-to-fill vacancies typically attract more job seeker applications, making screening the binding constraint on caseworker time. There is no direct effect via sourcing (panel c) for these vacancies.

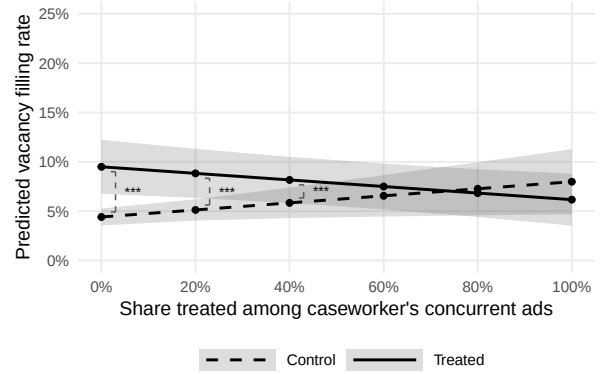
For hard-to-fill vacancies (right column), the picture is different. Panel (d) reveals a significant direct treatment effect on sourcing: treated vacancies are substantially more likely to result in a hire through caseworker-initiated referrals. This suggests that for vacancies where suitable candidates are scarce, the matching tool’s recommendations provide genuine value by helping caseworkers identify candidates they would not have found otherwise. The screening channel (panel f) also shows positive effects, though the spillover pattern is less pronounced than for easy-to-fill vacancies.

These findings suggest that the matching tool operates through distinct mechanisms depending on vacancy characteristics. For easy-to-fill vacancies, where candidate supply is abundant, the tool primarily generates time savings that caseworkers reallocate toward screening. For hard-to-fill vacancies, where finding suitable candidates is the core challenge, the tool directly improves sourcing by identifying matches that caseworkers might otherwise miss. The patterns are less pronounced but similar at the 90-day horizon (Table A10).

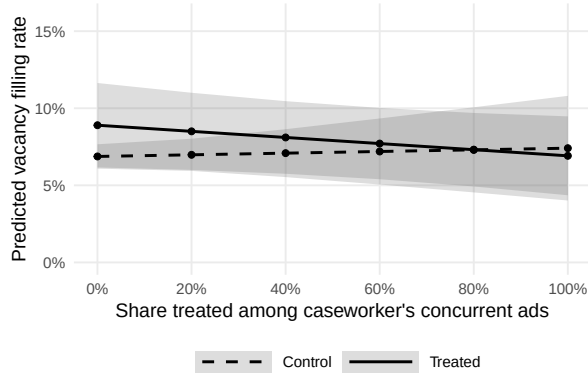
Figure 9: Predicted vacancy filling rates by treatment status and caseworker workload composition.



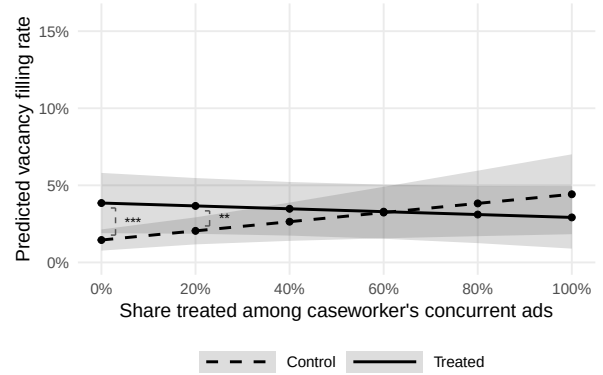
(a) All hires – Easy to fill



(b) All hires – Hard to fill



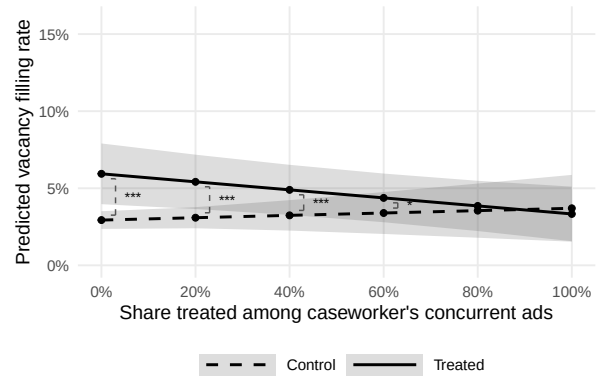
(c) Hires via sourcing – Easy to fill



(d) Hires via sourcing – Hard to fill



(e) Hires via screening – Easy to fill



(f) Hires via screening – Hard to fill

Notes: Predicted vacancy filling rates within 45 days based on the OLS estimates reported in Table 6. The x-axis shows the share of treated vacancies among the caseworker's concurrent workload, defined as the rolling sum of preselection-mode vacancies assigned to the same caseworker within a centered 7-day window around the vacancy's creation date, excluding the vacancy itself, controlling for the number of concurrent ads in the caseworkers portfolio. The average number of concurrent ads with preselection flag per 7-day window is 4.11. Predictions are computed at mean effect for all other covariates and fixed effects. "Easy to fill" and "Hard to fill" are defined by a median split on the IEE score (France Travail's prediction of the probability of filling the vacancy within 30 days). Shaded areas represent 95% confidence intervals. Significance stars indicate the statistical significance of the treatment effect: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 6: Treatment effect by vacancy filling difficulty and caseworker workload composition (45-day horizon).

	Easy to fill			Hard to fill		
	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Var.:	All	Sourcing	Screening	All	Sourcing	Screening
Treated	0.0547*** (0.0193)	0.0203 (0.0134)	0.0365** (0.0155)	0.0509*** (0.0132)	0.0240*** (0.0093)	0.0300*** (0.0096)
Concurrent ads same CW	-0.0016 (0.0012)	-4.18e-5 (0.0008)	-0.0016* (0.0009)	0.0001 (0.0009)	0.0007 (0.0007)	-0.0005 (0.0006)
Conc. ads with preselection flag same CW	-0.0014 (0.0020)	-0.0011 (0.0014)	-0.0002 (0.0015)	-0.0022 (0.0014)	-0.0017* (0.0010)	-0.0004 (0.0010)
Share treated among conc. ads with flag	0.0474* (0.0243)	0.0054 (0.0167)	0.0489** (0.0197)	0.0357** (0.0162)	0.0297** (0.0124)	0.0076 (0.0109)
Treated $\times$ Share treated	-0.0686** (0.0299)	-0.0252 (0.0211)	-0.0446* (0.0239)	-0.0690*** (0.0206)	-0.0390** (0.0156)	-0.0337** (0.0143)
Strata $\times$ sub-experiment FE	X	X	X	X	X	X
Agency $\times$ sub-experiment FE	X	X	X	X	X	X
Caseworker FE	X	X	X	X	X	X
Observations	6,847	6,847	6,847	6,564	6,564	6,564
R2	0.12398	0.12458	0.11639	0.10923	0.09878	0.10250
Control mean	0.1485	0.0709	0.0806	0.0646	0.0332	0.0319
Mean concurrent ads (preselection)	4.070	4.070	4.070	4.180	4.180	4.180

Notes: OLS regressions of vacancy filling outcomes (dummy for at least one hire within 45 days) on treatment status, share of treated ads among caseworker’s concurrent workload, and their interaction. Concurrent workload is measured as the rolling sum of preselection-mode vacancies assigned to the same caseworker within a centered 7-day window around the vacancy’s creation date, excluding the vacancy itself. “Easy to fill” and “Hard to fill” are defined by a median split on the IEE score (France Travail’s prediction of the probability of filling the vacancy within 30 days). All specifications include strata $\times$ sub-experiment, agency $\times$ sub-experiment, and caseworker fixed effects. Standard errors clustered at the establishment level in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.4 Congestion among treated ads

THIS SECTION SHOWS THAT IT IS NOT THAT YOU ALWAYS CONTACT THE SAME JOBSEEKER CAN STILL PUT IT SOMEWHERE ELSE EG MERGE WITH SECTION BEFORE

Table 7 shows that jobseeker level congestion, i.e that with many treated vacancies in a week the pool of workers gets exhausted, is not the reason for the lack of positive spillovers on vacancies that are themselves in the treatment group.

5.5 Survey evidence on caseworker practices

To complement the quantitative evidence on mechanisms, France Travail conducted a survey among caseworkers in agencies in which the experiment was deployed. Since the randomised was conducted at the establishment level rather than the caseworker level, all the caseworkers in those agencies can be thought of as having receiving treatment. The survey was administered in February 2025. Out of 86 respondents, 77 (88%) indicated that they have used MatchFT, 72 of them responded to questions about the usage. Full results are reported in Appendix B.1.

Caseworkers report that the tool changes how they identify candidates. Over half (57%) indicate they now propose vacancies to more candidates, and a similar share (54%) re-

Table 7: Candidate pool overlap across concurrent MatchFT requests.

	(1)	(2)
Dependent Var.:	Mean Jaccard index	Share of pairs with overlap
Deviation from expected treatment share	-0.0040 (0.0046)	0.0068 (0.0223)
Concurrent ads decile FE	X	X
Concurrent ads (preselection) decile FE	X	X
Strata <i>imes</i> sub-experiment FE	X	X
Agency <i>imes</i> sub-experiment FE	X	X
Caseworker FE	X	X
-----	-----	-----
Observations	3,073	3,073
R2	0.58587	0.43142
Within R2	0.00055	0.00010
Sample mean	0.0042	0.0262

Notes: OLS regressions of candidate pool overlap measures on the deviation of the realized treatment share among a caseworker's concurrent vacancies from the experimentally expected share (50% before February 6, 2025; 75% after). The unit of observation is a treated vacancy that received at least one MatchFT recommendation, restricted to vacancies for which the same caseworker handled at least one other treated vacancy within a centered 7-day window. For each focal vacancy i , all other treated vacancies j handled by the same caseworker within ± 3 days of vacancy i 's entry into preselection mode are identified, and the Jaccard similarity index is computed between the sets of candidates recommended by MatchFT for vacancy i and vacancy j , defined as the size of the intersection divided by the size of the union of the two candidate sets. Column (1) reports the mean Jaccard index across all concurrent vacancy pairs for the focal vacancy. Column (2) reports the share of concurrent vacancy pairs for which the two candidate sets have at least one candidate in common (i.e. Jaccard > 0). Both outcomes are aggregated to the focal vacancy level prior to estimation. All specifications include decile fixed effects for the number of concurrent vacancies (all modes) and for the number of concurrent preselection-mode vacancies, as well as strata $\times$ sub-experiment, agency $\times$ sub-experiment, and caseworker fixed effects. Standard errors clustered at the establishment level in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

port reaching *different* candidates than they would have otherwise. 87% report contacting candidates with sparser dossiers, 82% reach geographically more distant candidates, and 74% contact candidates with lower algorithmic matching scores. These patterns suggest that the tool helps caseworkers overcome information frictions that would otherwise lead them to overlook suitable candidates. Caseworkers engage actively with the tool's output: two-thirds read the AI-mediated conversations with candidates systematically, and 64% review all candidacies where any exchange occurred rather than limiting attention to those who passed all screening questions. Most caseworkers (89%) check responses within 24 hours and spend 1–5 minutes per candidate. While caseworkers call roughly 3 out of 10 responding candidates for follow-up—primarily to obtain additional profile information (91%) or provide details about the vacancy (84%)—the majority (55%) report that the tool does not lead them to make more calls overall.

When asked about the tool's positive features, caseworkers most frequently cite the ability to contact candidates via SMS (93%) and to reach more people simultaneously (87%). A majority also value faster candidate response times (74%) and the additional information gleaned from AI-mediated conversation details (68%). Notably, only 30% report higher response rates than other channels, suggesting the tool's value lies primarily in expanding reach and reducing friction rather than improving conversion rates per contact.

The caseworkers' answers to questions about the perceived benefits of the tool corroborate our regression findings. 56% of caseworkers report that the tool improved their time efficiency, compared to 49% reporting improved service quality for job seekers and 38% improved matching service quality for employers. The prominence of time savings—the most frequently cited improvement—is consistent with our spillover analysis showing that time freed on treated vacancies is reallocated toward screening tasks on other vacancies. Notably, only 4% report worsened service for job seekers and 1% for employers, suggesting that efficiency gains do not come at the cost of service quality. When asked about the types of vacancies where the tool is most useful, 93% cite vacancies with few or irrelevant spontaneous applications, while only 22% find it useful for vacancies attracting many candidates. This asymmetry mirrors our finding that the direct treatment effect on sourcing is concentrated among hard-to-fill vacancies, where candidate supply is scarce.

Table 8: Treatment effect on vacancy filling by caseworker tenure.

	Take-up	Hires via portal (45d)		
	(1)	(2)	(3)	(4)
Channel:	Tool use	All	Sourcing	Screening
New caseworker	0.0542** (0.0271)	-0.0121 (0.0179)	0.0148 (0.0149)	-0.0251** (0.0110)
Treated		0.0135** (0.0062)	0.0061 (0.0044)	0.0093** (0.0045)
Treated $\times$ New caseworker		0.0161 (0.0221)	-0.0192 (0.0171)	0.0315** (0.0152)
Strata $\times$ sub-experiment FE	X	X	X	X
Agency $\times$ sub-experiment FE	X	X	X	X
Observations	11,350	17,221	17,221	17,221
R2	0.13512	0.01847	0.01858	0.01625
Take-up (experienced)	0.3610			
Control mean for experienced		0.1103	0.0545	0.0576
% effect — experienced		12.2%	11.3%	16.1%
% effect — new caseworkers		30.8%	-18.9%	136%

Notes: Column (1) reports an OLS regression of tool take-up on an indicator for new caseworkers (tenure < 1 year), estimated on treated vacancies only, with strata $\times$ sub-experiment and agency $\times$ sub-experiment fixed effects. Columns (2)–(4) report OLS regressions of vacancy filling outcomes within 45 days on the treatment indicator (*Treated*), the new-caseworker indicator (*New caseworker*), and their interaction, with the same fixed effects. The reference category is a control vacancy handled by an experienced caseworker (tenure ≥ 1 year). Column (2) uses an indicator for at least one hire via the portal from any channel; column (3) restricts to hires via sourcing (caseworker-initiated referrals); column (4) restricts to hires via screening (job-seeker-initiated referrals approved by the caseworker). The unit of observation is one vacancy posting. The table also reports effects translated into a percentage of the relevant control mean: for experienced caseworkers the numerator is the Treated coefficient; for new caseworkers it is the sum of the Treated and Treated $\times$ New caseworker coefficients. Standard errors clustered at the establishment level in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.6 Caseworker tenure

Table 8 investigates whether this effect differs by caseworker tenure. The treatment effect on overall hiring is XXXpp for experienced caseworkers and YYyp for new caseworkers, with the interaction term being significant at the 5% level. - Much larger effect for new caseworkers - Much lower baseline for new cw

TDODO add 90 day horizon tables

Because new caseworkers constitute a small subgroup of the analysis sample small cell sizes could potentially produce chance imbalances even under valid randomization. Table A12 in the appendix examines whether firm and ad characteristics differ between treated and control ads within the two caseworker groups. The table shows that almost all characteristics are balanced also within the subgroups TODO

Table 9: Treatment or spillover effect on vacancy filling by caseworker tenure.

	Take-up	Hires via portal (45d)		
	(1)	(2)	(3)	(4)
Channel:	Tool use	All	Sourcing	Screening
New caseworker	0.0542** (0.0271)	-0.0254 (0.0189)	0.0053 (0.0159)	-0.0285** (0.0111)
Treated or high-spillover		0.0158** (0.0064)	0.0053 (0.0046)	0.0125*** (0.0047)
Treated or high-spillover $\times$ New caseworker		0.0297 (0.0231)	-0.0075 (0.0183)	0.0328** (0.0151)
Strata $\times$ sub-experiment FE	X	X	X	X
Agency $\times$ sub-experiment FE	X	X	X	X
Concurrent ads quintile FE		X	X	X
Concurrent ads (presel.) quintile FE		X	X	X
-----	-----	-----	-----	-----
Observations	11,350	17,091	17,091	17,091
R2	0.13512	0.02074	0.02027	0.01755
Take-up (experienced)	0.3610			
Control mean for experienced		0.1113	0.0549	0.0582

Notes: Column (1) reports an OLS regression of tool take-up on an indicator for new caseworkers (tenure < 1 year), estimated on treated vacancies only, with strata $\times$ sub-experiment and agency $\times$ sub-experiment fixed effects. Columns (2)–(4) report OLS regressions of vacancy filling outcomes within 45 days on the treatment-or-spillover indicator (*Treated or high-spillover*), the new-caseworker indicator (*New caseworker*), and their interaction, with the same fixed effects and additional quintile fixed effects for the total number of concurrent vacancies and for the number of concurrent vacancies in preselection mode. A vacancy is classified as high-spillover if the share of treated vacancies among the caseworker’s concurrent caseload falls in the top tercile, computed separately for the periods before and after the treatment share increase on February 6, 2025. The reference category is a control vacancy not in the high-spillover tercile, handled by an experienced caseworker (tenure ≥ 1 year). Column (2) uses an indicator for at least one hire via the portal from any channel; column (3) restricts to hires via sourcing (caseworker-initiated referrals); column (4) restricts to hires via screening (job-seeker-initiated referrals approved by the caseworker). The unit of observation is one vacancy posting. The table also reports effects translated into a percentage of the relevant control mean: for experienced caseworkers the numerator is the Treated or high-spillover coefficient; for new caseworkers it is the sum of the Treated or high-spillover and Treated or high-spillover $\times$ New caseworker coefficients. Standard errors clustered at the establishment level in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.7 Caseworker quality

Tenure is a coarse proxy for how good a caseworker is at filling vacancies. In this subsection we construct a direct measure of caseworker quality, ask whether it predicts who adopts the tool, verify that the treatment effect is unaffected by it, and finally use it to put the size of the treatment effect on a scale that has an interpretation.

We measure caseworker value added (CWVA) on the vacancies of the year preceding the experiment, from 21 November 2023 to 20 November 2024, a period that is disjoint from the experimental sample in which the measure is later used. On that pre-period sample we estimate two linear probability models of the same outcome we study in the experiment — whether the vacancy is filled by a referred job seeker within a given horizon — one including caseworker fixed effects and one without:

$$y_i = \alpha_{j(i)} + \beta s_i + \delta_{a(i)} + \gamma_{m(i)} + \varepsilon_i, \quad (2)$$

$$y_i = \tilde{\beta} s_i + \tilde{\delta}_{a(i)} + \tilde{\gamma}_{m(i)} + u_i, \quad (3)$$

where y_i is filling of vacancy i , s_i its IEE score, and $\delta_{a(i)}$ and $\gamma_{m(i)}$ agency and calendar-month fixed effects. Vacancies whose IEE score is not computed are dropped. Caseworker j 's value added is the difference between the two models' predictions, averaged over j 's pre-period vacancies,

$$\text{CWVA}_j = \mathbb{E}[y_i \mid \alpha_j, X_i] - \mathbb{E}[y_i \mid X_i], \quad (4)$$

which is expressed in probability points and answers the question: how much more often than an observationally identical vacancy in the same agency and month does this caseworker fill a vacancy? The construction is in the spirit of the leniency-design literature (Kolesár, 2013; Goldsmith-Pinkham et al., 2025): as there, the object of interest is a decision-maker fixed effect that must be estimated on data other than the observations it is later applied to, so that the sampling error in a caseworker's own average does not mechanically correlate with the outcome being explained. We do not, however, use it as an instrument.

We keep the 600 of 665 caseworkers with at least five usable pre-period vacancies, a median of 46, which covers 93.5% of the experimental vacancies. The measure is winsorised at the 1st and 99th percentile and divided by its standard deviation across caseworkers, so that a unit corresponds to one standard deviation of caseworker value added. Because the outcome accumulates over time, we construct one measure per horizon and always pair the h -day measure with the h -day outcome. Figure A3 in the appendix plots the

distribution: the gap between a 90th- and a 10th-percentile caseworker is 16.8 percentage points of filling probability. That figure is the spread of the *estimated* value added and therefore still contains the sampling noise in a caseworker’s pre-period average; the disattenuated magnitude we quote below is smaller.

Column (1) of Table 10 regresses, at the caseworker level, an indicator for ever having sent a *MatchFT* request on caseworker value added. Caseworkers who were never assigned a treated vacancy had no adoption decision to make and are excluded. Take-up is not random across caseworkers: a one standard deviation higher pre-experiment value added is associated with a 5.9 percentage point lower probability of ever using the tool. The gradient is precisely estimated but modest against a base take-up rate of 91%, and value added accounts for only 3.5% of the variation in adoption across caseworkers.

Columns (2) and (3) add caseworker value added to the baseline specification on the same sample. The treatment effect moves from 1.49 to 1.45 percentage points, well within a standard error. Table A13 in the appendix repeats the exercise at every horizon and shows the same stability throughout. This is what randomisation should deliver — treated vacancies were not handed to better caseworkers — and it confirms that the effect we estimate is not a caseworker-composition artefact.

Value added is nonetheless a strong predictor of filling in the experimental period: a one standard deviation higher CWVA raises the filling probability by 2.04 percentage points at 45 days, against a control mean of 10.9%. Because the measure is estimated on a disjoint period, this coefficient is already net of the sampling noise that inflates the raw dispersion reported above. It gives the treatment effect a scale: assigning the tool to a vacancy is worth about 0.71 standard deviations of caseworker value added. Equivalently, the 90th–10th percentile caseworker gap is 2.17 standard deviations, or 4.4 percentage points of filling probability, and the treatment effect amounts to roughly a third of it. Handing a caseworker the tool moves a vacancy about as far as moving it a third of the way up the distribution of caseworker quality.

Column (4) splits the treatment effect by whether the caseworker ever used the tool, again among caseworkers with at least one treated vacancy. Never-users are a distinctive group: across the experiment they are 70 of the 603 caseworkers who were assigned at least one treated vacancy, they handle only 3.5% of the vacancies, carry a median of 12 treated vacancies against 29 for users, and have a value added 0.54 standard deviations above the mean — consistent with the adoption gradient documented above. Their treated vacancies show no statistically significant effect. Given how few such caseworkers there are, however, this is far from a precise zero: the confidence interval on their estimate runs from –2.5 to 11.1 percentage points and so excludes neither zero nor an

effect several times the one we estimate for users. We report the split for completeness rather than as a placebo test the design can support.

Table 10: Tool usage, vacancy filling and caseworker value added

	Take-up	Vacancy filling		
	(1)	(2)	(3)	(4)
Outcome:	Used MatchFT	Within 45 days	Within 45 days	Within 45 days
Constant	0.9094*** (0.0127)			
Caseworker value added (SD)	-0.0589*** (0.0210)		0.0204*** (0.0043)	
Treated		0.0149** (0.0062)	0.0145** (0.0062)	
Treated $\times$ MatchFT-user CW				0.0136** (0.0063)
Treated $\times$ never-user CW				0.0432 (0.0346)
Strata $\times$ sub-experiment FE		X	X	X
Agency $\times$ sub-experiment FE		X	X	X
CW group FE				X
Unit	Caseworker	Job ad	Job ad	Job ad
Observations	499	16,103	16,103	16,082
R2	0.03470	0.02111	0.02328	0.02134
Take-up share	0.910			
Control mean		0.109	0.109	0.109
CWVA $p_{90} - p_{10}$ (SD)			2.17	

Notes: Caseworker value added is estimated on the vacancies of the year preceding the experiment (21 November 2023 to 20 November 2024) following equation (4), and is expressed in standard deviations of its distribution across caseworkers. Column (1) is estimated at the caseworker level: the dependent variable is an indicator for the caseworker ever having sent a *MatchFT* request, and the sample is the caseworkers with at least one treated vacancy and a value added, so that caseworkers without an adoption decision to make are excluded. Standard errors are heteroskedasticity-robust. Columns (2) to (4) are estimated at the vacancy level on the experimental sample restricted to vacancies whose caseworker has a value added, with strata $\times$ sub-experiment and agency $\times$ sub-experiment fixed effects. The dependent variable is an indicator for the vacancy being filled by a referred job seeker within 45 days. Column (3) adds caseworker value added as a control; column (4) interacts treatment with an indicator for whether the caseworker ever used the tool, with caseworker-group fixed effects absorbing the main effect, so that each group's treatment effect is identified against its own control vacancies. CWVA $p_{90} - p_{10}$ is the spread of value added across the caseworkers of column (3), in standard deviations. Standard errors clustered at the establishment level in parentheses in columns (2) to (4). Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.8 Matching

Table 11: Treatment effect on the alignment between the candidate and the job posting.

	All candidates			Sourcing			Screening		
	N	Control mean	Coef. (q-value)	N	Control mean	Coef. (q-value)	N	Control mean	Coef. (q-value)
Job title match	134,495	0.145	0.002 (0.699)	38,419	0.221	0.002 (0.831)	96,076	0.115	0.000 (0.982)
Distance (log km)	126,614	2.626	-0.030 (0.240)	36,404	2.365	-0.018 (0.502)	90,210	2.727	-0.030 (0.336)
Meets experience requirement	122,192	0.859	0.006 (0.394)	37,210	0.888	0.006 (0.502)	84,982	0.846	0.006 (0.554)
Meets education requirement	49,194	0.353	0.004 (0.662)	13,497	0.396	0.010 (0.564)	35,697	0.337	-0.000 (0.982)
Meets driver's licence requirement	119,414	0.998	0.000 (0.662)	34,538	0.998	0.000 (0.741)	84,876	0.999	0.000 (0.702)
Contract type matches	134,495	0.541	-0.006 (0.662)	38,419	0.596	-0.014 (0.502)	96,076	0.519	-0.004 (0.982)
Working hours compatible	134,495	0.833	-0.006 (0.240)	38,419	0.879	0.010 (0.138)	96,076	0.815	-0.014*** (0.006)
Log absolute wage gap	29,997	0.322	-0.043* (0.054)	9,791	0.308	-0.047 (0.138)	20,204	0.329	-0.047** (0.030)

Notes: OLS regressions of match quality indicators on the treatment indicator, with strata $\times$ sub-experiment and agency $\times$ sub-experiment fixed effects. Each row is a separate regression. The unit of observation is a caseworker referral. Column (1) pools all referrals. Column (2) restricts to caseworker-initiated referrals (sourcing channel). Column (3) restricts to job-seeker-initiated referrals approved by the caseworker (screening channel). Match quality indicators are constructed by linking each referral to the job seeker's stated preferences and the vacancy requirements. Occupation match equals one if the job seeker's desired occupation matches the vacancy's posted occupation title. Experience sufficiency equals one if the job seeker's reported experience in the relevant occupation meets or exceeds the vacancy's minimum requirement; job seekers with no experience profile on record are treated as not meeting the requirement when the vacancy requires experience. Education match is defined only for vacancies that specify an education requirement and equals one if the job seeker's education level meets it. Licence match equals one if the vacancy imposes no mandatory driving licence requirement, or if it does and the job seeker holds the required category. Contract match equals one if the job seeker has no contract type preference, or if their preferred contract type matches the vacancy. The wage gap is the absolute difference in logs between the vacancy's posted maximum wage and the job seeker's stated reservation wage, both converted to an annual full-time equivalent basis; a negative coefficient indicates that treatment reduces the wage gap between vacancy and job seeker. Wage gap regressions exclude observations where either wage is missing or implausibly low. Standard errors clustered at the establishment level. q -values (in parentheses) are Benjamini–Hochberg adjusted p -values, computed separately within each channel across the eight outcomes. Stars denote significance based on q -values: *** $q < 0.01$, ** $q < 0.05$, * $q < 0.10$. See Table A11 for a detailed decomposition of each match dimension.

Table 12: Heterogeneous treatment effects on match quality, by caseworker tenure.

	All candidates				Sourcing				Screening			
	N	Ctrl. mean (exp. CW)	Treat (exp. CW)	Treat × new CW (<i>q</i> -value below)	N	Ctrl. mean (exp. CW)	Treat (exp. CW)	Treat × new CW (<i>q</i> -value below)	N	Ctrl. mean (exp. CW)	Treat (exp. CW)	Treat × new CW (<i>q</i> -value below)
Job title match	111,108	0.143	0.005 (0.523)	-0.046 (0.373)	31,765	0.215	0.009 (0.534)	-0.069 (0.470)	79,343	0.115	0.002 (0.852)	-0.028 (0.817)
Distance (log km)	104,476	2.621	-0.029 (0.377)	0.001 (0.988)	30,142	2.347	-0.013 (0.594)	-0.053 (0.627)	74,334	2.727	-0.030 (0.505)	-0.004 (0.958)
Meets experience requirement	100,880	0.860	0.007 (0.434)	-0.019 (0.730)	30,773	0.890	0.006 (0.534)	-0.000 (0.995)	70,107	0.847	0.006 (0.736)	-0.020 (0.817)
Meets education requirement	40,809	0.357	0.004 (0.661)	-0.038 (0.730)	11,076	0.399	0.009 (0.594)	-0.061 (0.577)	29,732	0.342	-0.002 (0.852)	-0.008 (0.958)
Meets driver's licence requirement	99,210	0.998	0.001 (0.523)	-0.001 (0.730)	28,678	0.998	0.001 (0.534)	-0.001 (0.627)	70,532	0.999	0.000 (0.736)	-0.001 (0.817)
Contract type matches	111,108	0.536	0.003 (0.840)	0.008 (0.981)	31,765	0.585	0.001 (0.932)	-0.054 (0.627)	79,343	0.517	0.004 (0.852)	0.033 (0.817)
Working hours compatible	111,108	0.832	-0.006 (0.377)	-0.017 (0.730)	31,765	0.876	0.011 (0.362)	-0.026 (0.511)	79,343	0.816	-0.014** (0.022)	-0.013 (0.817)
Log absolute wage gap	25,625	0.321	-0.039 (0.172)	0.032 (0.816)	8,242	0.305	-0.038 (0.362)	-0.051 (0.627)	17,378	0.329	-0.045* (0.090)	0.070 (0.817)
Application rank (relative)									79,343	0.532	-0.002 (0.736)	-0.001 (0.958)

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (stars based on *q*-values, BH correction applied separately to main-effect and interaction-term *p*-values within each channel).

New CW = tenure < 1 year at experiment start (2024-11-21). Reference category: experienced CW (≥ 1 year).

Notes: OLS regressions of match quality indicators on the treatment indicator, its interaction with an indicator for new caseworkers, and the new-caseworker indicator itself, with strata $\times$ sub-experiment and agency $\times$ sub-experiment fixed effects. Each row is a separate regression. The unit of observation is a caseworker referral. Column (1) pools all referrals. Column (2) restricts to caseworker-initiated referrals (sourcing channel). Column (3) restricts to job-seeker-initiated referrals approved by the caseworker (screening channel). “Treat (exp. CW)” is the treatment effect for experienced caseworkers (reference category). “Treat $\times$ new CW” is the differential treatment effect for new caseworkers, i.e. the sum of the two coefficients gives the treatment effect for new caseworkers. A caseworker is classified as new if their contract start date is less than one year before the experiment start date (2024-11-21); experienced caseworkers have at least one year of tenure at that date. Observations with missing tenure information are excluded. Control means are computed for the pure control group among experienced caseworkers. Match quality indicators are constructed by linking each referral to the job seeker’s stated preferences and the vacancy requirements; see Table 11 for definitions of each indicator. Standard errors clustered at the establishment level. *q*-values (in parentheses) are Benjamini–Hochberg adjusted *p*-values, computed separately within each channel: once across the eight main-effect coefficients and once across the eight interaction coefficients. Stars denote significance based on *q*-values: *** $q < 0.01$, ** $q < 0.05$, * $q < 0.10$.

Figure 10: Screening interface: comparison of applicant and ad characteristics.

Comparer l'offre et le profil du candidat		
	 Offre n°205JRRJ Voir le détail de l'offre	 Profil de M. [REDACTED] Voir le détail du profil
Intitulé du poste	Encadrant(e) technique d'insertion socioprofessionnelle	Pas de correspondance dans le profil
Lieu de travail	28 - CHARTRES	BAILLEAU-ARMENONVILLE (28) + 30 km
Expérience	3 ans d'expérience exigée - encadrement en ACI	Pas de correspondance dans le profil
Formation(s)	Bac ou équivalent - Encadrement technique insertion exigé	Pas de correspondance dans le profil
Permis	B - Véhicule léger exigé	B - Véhicule léger
Type de contrat	CDI (CDI)	CDI (Contrat à durée indéterminée)
Durée hebdomadaire	39 heures	Temps plein
Salaire	2300€ à 2500€ Mensuel, sur 12 mois Complémentaire santé, CSE	1900€ mensuel

Notes: This screenshot from the France Travail caseworker interface shows the screen a caseworker sees when assessing a jobseeker-initiated application and deciding whether to forward it to the firm.

References

- ABEL, M., R. BURGER, AND P. PIRAINO (2020): “The Value of Reference Letters: Experimental Evidence from South Africa,” *American Economic Journal: Applied Economics*, 12, 40–71.
- ALGAN, Y., B. CRÉPON, AND D. GLOVER (2022): “Are Active Labor Market Policies Directed at Firms Effective? Evidence from a Randomized Evaluation with Local Employment Agencies,” .
- ALTMANN, S., A. FALK, S. JÄGER, AND F. ZIMMERMANN (2018): “Learning about job search: A field experiment with job seekers in Germany,” *Journal of Public Economics*, 164, 33–49.
- ALTMANN, S., A. GLENNY, R. MAHLSTEDT, AND A. SEBALD (2023): “The Direct and Indirect Effects of Online Job Search Advice,” .
- BEHAGHEL, L., S. DROMUNDO, M. GURGAND, Y. HAZARD, AND T. ZUBER (2024): “The Potential of Recommender Systems for Directing Job Search: A Large-Scale Experiment,” *SSRN Electronic Journal*.
- BELOT, M., P. KIRCHER, AND P. MULLER (2022): “How Wage Announcements Affect Job Search—A Field Experiment,” *American Economic Journal: Macroeconomics*, 14, 1–67.
- BIASI, B. (2021): “The Labor Market for Teachers under Different Pay Schemes,” *American Economic Journal: Economic Policy*, 13, 63–102.
- BIED, G., P. CAILLOU, B. CRÉPON, E. PÉRENNES, AND M. SEBAG (2023): “Designing Labor Market Recommender Systems: How to Improve Human-based Search,” .
- BUGNI, F. A., I. A. CANAY, AND A. M. SHAIKH (2018): “Inference Under Covariate-Adaptive Randomization,” *Journal of the American Statistical Association*, 113, 1784–1796.
- CARD, D., J. KLUVE, AND A. WEBER (2018): “What Works? A Meta Analysis of Recent Active Labor Market Program Evaluations,” *Journal of the European Economic Association*, 16, 894–931.
- CARRANZA, E., R. GARLICK, K. ORKIN, AND N. RANKIN (2022): “Job Search and Hiring with Limited Information about Workseekers’ Skills,” *American Economic Review*, 112, 3547–3583.
- CEDERLÖF, J., M. SÖDERSTRÖM, AND J. VIKSTRÖM (2025): “The Role of Caseworkers: Job Finding, Job Quality and Determinants of Value-Added,” *Journal of the European Economic Association*, jvaf055.
- CHEUNG, M., J. EGEBAK, A. FORSLUND, L. LAUN, M. RÖDIN, AND J. VIKSTRÖM (2025): “Does Job Search Assistance Reduce Unemployment? Evidence on Displacement Effects and Mechanisms,” *Journal of Labor Economics*, 43, 47–81.

- CRÉPON, B., E. DUFLO, M. GURGAND, R. RATHELOT, AND P. ZAMORA (2013): “Do Labor Market Policies Have Displacement Effects? Evidence from a Clustered Randomized Experiment,” *The Quarterly Journal of Economics*, 128, 531–580.
- DAL BÓ, E., F. FINAN, AND M. A. ROSSI (2013): “Strengthening State Capabilities: The Role of Financial Incentives in the Call to Public Service,” *The Quarterly Journal of Economics*, 128, 1169–1218.
- GAUTIER, P., P. MULLER, B. VAN DER KLAUW, M. ROSHOLM, AND M. SVARER (2018): “Estimating Equilibrium Effects of Job Search Assistance,” *Journal of Labor Economics*, 36, 1073–1125.
- GOLDSMITH-PINKHAM, P., P. HULL, AND M. KOLESÁR (2025): “Leniency Designs: An Operator’s Manual,” Working paper, National Bureau of Economic Research.
- HOFFMAN, M., L. B. KAHN, AND D. LI (2018): “Discretion in Hiring,” *The Quarterly Journal of Economics*, 133, 765–800.
- KOLESÁR, M. (2013): “Estimation in an instrumental variables model with treatment effect heterogeneity,” Working paper.
- KROFT, K., F. LANGE, AND M. J. NOTOWIDIGDO (2013): “Duration Dependence and Labor Market Conditions: Evidence from a Field Experiment,” *The Quarterly Journal of Economics*, 128, 1123–1167.
- LE BARBANCHON, T., L. HENSVIK, AND R. RATHELOT (2023): “How Can AI Improve Search and Matching? Evidence From 59 Million Personalized Job Recommendations,” .
- MUÑOZ, P. AND C. OTERO (2025): “Managers and Public Hospital Performance,” *American Economic Review*, 115, 4040–4074.
- MUÑOZ, P. AND M. PREM (2024): “Managers’ Productivity and Recruitment in the Public Sector,” *American Economic Journal: Economic Policy*, 16, 223–253.
- PALLAIS, A. (2014): “Inefficient Hiring in Entry-Level Labor Markets,” *American Economic Review*, 104, 3565–3599.
- SCHIPROWSKI, A. (2020): “The Role of Caseworkers in Unemployment Insurance: Evidence from Unplanned Absences,” *Journal of Labor Economics*, 38, 1189–1225.
- TRIFONOV, J., Y. BAI, A. SHAIKH, AND M. TABORD-MEEHAN (2024): “sreg: Stratified Randomized Experiments,” .

Figure A1: Screening interface: comparison of applicant and ad characteristics.

Comparer l'offre et le profil du candidat		
	 Offre n°205JRRJ Voir le détail de l'offre	 Profil de M. [REDACTED] Voir le détail du profil
Intitulé du poste	Encadrant(e) technique d'insertion socioprofessionnelle	Pas de correspondance dans le profil
Lieu de travail	28 - CHARTRES	BAILLEAU-ARMENONVILLE (28) + 30 km
Expérience	3 ans d'expérience exigée - encadrement en ACI	Pas de correspondance dans le profil
Formation(s)	Bac ou équivalent - Encadrement technique insertion exigé	Pas de correspondance dans le profil
Permis	B - Véhicule léger exigé	B - Véhicule léger
Type de contrat	CDI (CDI)	CDI (Contrat à durée indéterminée)
Durée hebdomadaire	39 heures	Temps plein
Salaire	2300€ à 2500€ Mensuel, sur 12 mois Complémentaire santé, CSE	1900€ mensuel

Notes: This screenshot from the France Travail caseworker interface shows the screen a caseworker sees when assessing a jobseeker-initiated application and deciding whether to forward it to the firm.

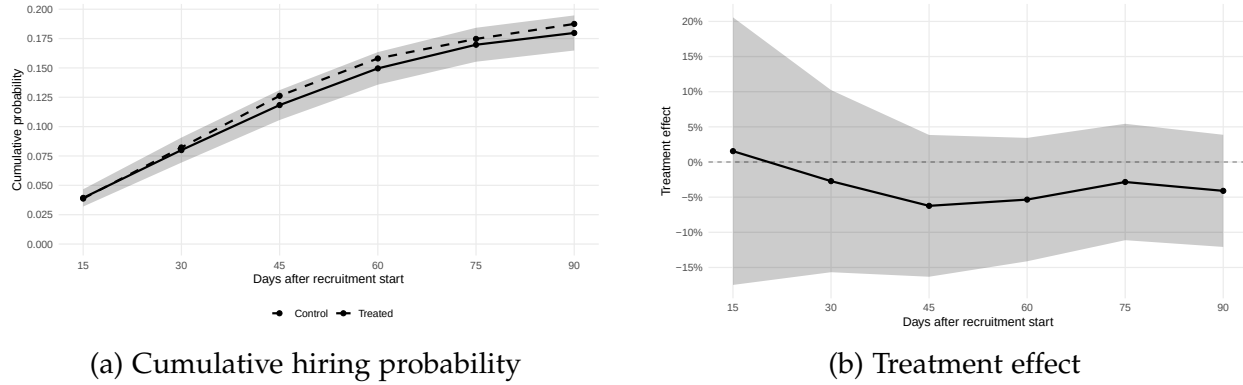
Table A1: Robustness: Strata Fixed Effects Only

Vacancy filled						
	(1)	(2)	(3)	(4)	(5)	(6)
1 Horizon:	15 days	30 days	45 days	60 days	75 days	90 days
2						
3 Treated	0.0027 (0.0032)	0.0092* (0.0050)	0.0137** (0.0061)	0.0153** (0.0068)	0.0166** (0.0072)	0.0134* (0.0075)
5 Strata × sub-experiment FE	X	X	X	X	X	X
6						
7 Observations	17,225	17,225	17,225	17,225	17,225	17,225
8 R2	0.00128	0.00198	0.00230	0.00254	0.00385	0.00445
9 Control mean	0.0316703558658267	0.0727056019070322	0.109654350417163	0.138430103865146	0.159884215903286	0.176570747488507

Notes: OLS regressions of vacancy filling outcomes on treatment status. Specifications include strata × sub-experiment fixed effects only (no agency fixed effects). Standard errors clustered at the establishment level in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

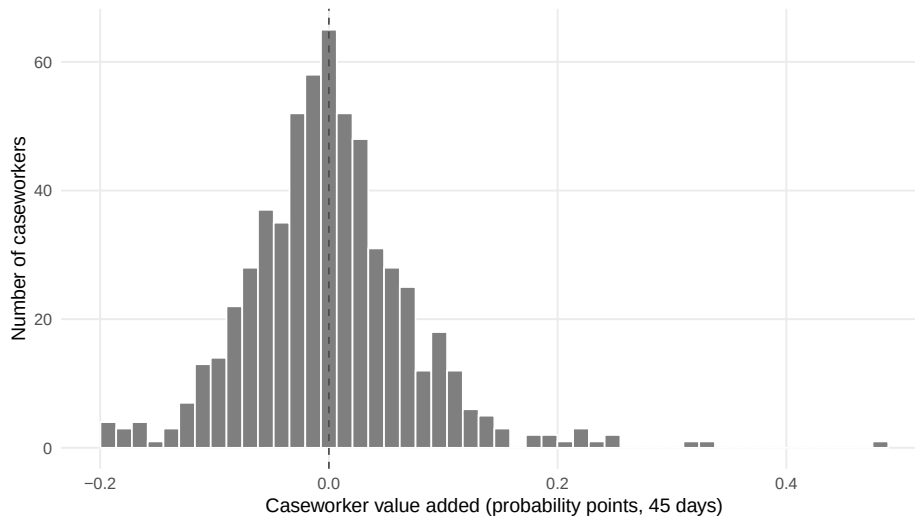
A Extra Figures

Figure A2: Placebo Test: Pre-Treatment Period



Notes: Placebo test using vacancies from November 2023 to April 2024, before the experiment began. Treatment status is assigned based on the establishment’s eventual treatment assignment. Panel (a) shows cumulative hiring probabilities for “treated” and “control” vacancies. Panel (b) shows the placebo treatment effect as a percentage of the control mean. Estimates are based on stratified randomization inference with strata $\times$ sub-experiment fixed effects. Shaded areas and error bars represent 95% confidence intervals.

Figure A3: Distribution of caseworker value added



Notes: One observation is one caseworker. Caseworker value added is estimated on the vacancies of the year preceding the experiment (21 November 2023 to 20 November 2024) following equation (4), as the difference between the predicted filling probability within 45 days with and without caseworker fixed effects, averaged over the caseworker’s pre-period vacancies, and is shown here in probability points. The sample is the 600 caseworkers with at least five usable pre-period vacancies, a median of 46. The dashed line marks zero, the value added of a caseworker who fills vacancies exactly as often as predicted by the agency, the calendar month and the IEE score. The dispersion shown includes the sampling error in a caseworker’s pre-period average and therefore overstates the dispersion of true value added; see Section 5.7.

Table A2: Robustness: Agency Fixed Effects

		Vacancy filled					
		(1)	(2)	(3)	(4)	(5)	(6)
1	Horizon:	15 days	30 days	45 days	60 days	75 days	90 days
2							
3	Treated	0.0038 (0.0032)	0.0099** (0.0049)	0.0144** (0.0060)	0.0156** (0.0066)	0.0172** (0.0070)	0.0130* (0.0073)
5	Strata $\times$ sub-experiment FE	X	X	X	X	X	X
6	Agency $\times$ sub-experiment FE	X	X	X	X	X	X
7							
8	Observations	17,225	17,225	17,225	17,225	17,225	17,225
9	R2	0.01650	0.01749	0.01842	0.02216	0.02555	0.02618
10	Control mean	0.0316703558658267	0.0727056019070322	0.109654350417163	0.138430103865146	0.159884215903286	0.176570747488507

Notes: OLS regressions of vacancy filling outcomes on treatment status. Specifications include strata $\times$ sub-experiment and agency $\times$ sub-experiment fixed effects. Standard errors clustered at the establishment level in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A3: Robustness: Additional Controls

		Vacancy filled					
		(1)	(2)	(3)	(4)	(5)	(6)
1	Horizon:	15 days	30 days	45 days	60 days	75 days	90 days
2							
3	Treated	0.0025 (0.0032)	0.0078* (0.0048)	0.0121** (0.0057)	0.0133** (0.0063)	0.0149** (0.0067)	0.0107 (0.0070)
5	Strata $\times$ sub-experiment FE	X	X	X	X	X	X
6	Agency $\times$ sub-experiment FE	X	X	X	X	X	X
7	Controls	X	X	X	X	X	X
8							
9	Observations	17,225	17,225	17,225	17,225	17,225	17,225
10	R2	0.04292	0.06040	0.06161	0.06682	0.07107	0.07122
11	Control mean	0.0316703558658267	0.0727056019070322	0.109654350417163	0.138430103865146	0.159884215903286	0.176570747488507

Notes: OLS regressions of vacancy filling outcomes on treatment status, controlling for the number of positions advertised and pre-experiment hiring quintiles. Specifications include strata $\times$ sub-experiment and agency $\times$ sub-experiment fixed effects. Standard errors clustered at the establishment level in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A4: Robustness: PPML Specification

		Number of hires via referral					
		(1)	(2)	(3)	(4)	(5)	(6)
1	Horizon:	15 days	30 days	45 days	60 days	75 days	90 days
2							
3	Treated	0.1499 (0.1169)	0.1393* (0.0815)	0.1132* (0.0664)	0.0978 (0.0608)	0.0633 (0.0574)	0.0297 (0.0549)
4	Number of positions (log)	0.9366*** (0.1894)	0.6435*** (0.1520)	0.6828*** (0.1293)	0.6230*** (0.1196)	0.6674*** (0.1134)	0.6820*** (0.1038)
6	Strata $\times$ sub-experiment FE	X	X	X	X	X	X
7	Agency $\times$ sub-experiment FE	X	X	X	X	X	X
8							
9	Observations	16,214	17,121	17,202	17,225	17,225	17,225
10	Control mean	0.0316703558658267	0.0727056019070322	0.109654350417163	0.138430103865146	0.159884215903286	0.176570747488507

Notes: PPML regressions of the number of hires via referral on treatment status. Specifications include strata $\times$ sub-experiment and agency $\times$ sub-experiment fixed effects and control for the log number of positions advertised. Standard errors clustered at the establishment level in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A5: Robustness: All Positions Filled Outcome

		All positions filled via referral					
		(1)	(2)	(3)	(4)	(5)	(6)
1	Horizon:	15 days	30 days	45 days	60 days	75 days	90 days
2							
3	Treated	0.0047 (0.0031)	0.0119** (0.0048)	0.0160*** (0.0059)	0.0173*** (0.0065)	0.0184*** (0.0069)	0.0143** (0.0072)
5	Strata $\times$ sub-experiment FE	X	X	X	X	X	X
6	Agency $\times$ sub-experiment FE	X	X	X	X	X	X
7							
8	Observations	17,225	17,225	17,225	17,225	17,225	17,225
9	R2	0.01639	0.01740	0.01859	0.02239	0.02575	0.02632
10	Control mean	0.0316703558658267	0.0727056019070322	0.109654350417163	0.138430103865146	0.159884215903286	0.176570747488507

Notes: OLS regressions where the outcome is an indicator for whether all advertised positions were filled through caseworker referrals within the specified horizon. Specifications include strata $\times$ sub-experiment and agency $\times$ sub-experiment fixed effects. Standard errors clustered at the establishment level in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A6: Balance test: Pre-determined covariates across treatment intensity categories.

	Pred. filling rate		Firm's pre-experiment hires	
	(1)	(2)	(3)	(4)
Dependent Var.:	IEE Score	All	Sourcing	Screening
Control $\times$ Few treated	.	.	.	.
Control $\times$ Medium	-0.0058 (0.0062)	-0.2566** (0.1155)	-0.3323*** (0.1222)	-0.1586 (0.1547)
Control $\times$ Many treated	0.0048 (0.0067)	-0.0342 (0.1117)	0.0160 (0.1404)	-0.0734 (0.1225)
Treated $\times$ Few treated	0.0094 (0.0058)	-0.0108 (0.1334)	-0.0950 (0.1477)	0.1017 (0.1566)
Treated $\times$ Medium	0.0045 (0.0062)	-0.0923 (0.1169)	-0.1692 (0.1344)	0.0225 (0.1382)
Treated $\times$ Many treated	0.0061 (0.0060)	-0.1279 (0.1112)	-0.1796 (0.1295)	-0.0458 (0.1370)
catcwadsother	X	X	X	X
catcwadsother	X	X	X	X
Strata $\times$ sub-experiment FE	X	X	X	X
Agency $\times$ sub-experiment FE	X	X	X	X
Caseworker FE	X	X	X	X
Family	OLS	Poisson	Poisson	Poisson
Observations	13,511	13,785	13,066	12,243
R2	0.23113	—	—	—
Control mean	0.463	0.874	0.498	0.377
Mean concurrent ads (preselection)	4.150	4.150	4.150	4.150

Notes: Each observation is a vacancy in preselection mode during the experimental period. The dependent variable in column (1) is the IEE score, a prediction of the probability of filling the vacancy within 30 days computed by France Travail (restricted to vacancies with a non-missing score). Columns (2)–(4) report Poisson regressions of the establishment's total hires following a referral in the year before the experiment (September 2023–August 2024), decomposed by channel. The reference category is control vacancies in the lowest tercile of treatment intensity among concurrent vacancies. All specifications include strata $\times$ sub-experiment, agency $\times$ sub-experiment, and caseworker fixed effects, and control for the number of concurrent vacancies. Standard errors clustered at the establishment level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Treatment effect by caseworker workload composition

	Filled within 45 days			Filled within 90 days		
	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Var.:	All	Sourcing	Screening	All	Sourcing	Screening
Control × Few treated	.	.	.	.	.	.
Control × Medium	0.0179 (0.0118)	0.0116 (0.0088)	0.0097 (0.0084)	0.0086 (0.0143)	0.0062 (0.0104)	0.0062 (0.0107)
Control × Many treated	0.0284** (0.0129)	0.0047 (0.0090)	0.0263*** (0.0098)	0.0217 (0.0154)	0.0027 (0.0115)	0.0212* (0.0119)
Treated × Few treated	0.0386*** (0.0101)	0.0171** (0.0072)	0.0244*** (0.0077)	0.0381*** (0.0123)	0.0235** (0.0093)	0.0184** (0.0092)
Treated × Medium	0.0340*** (0.0102)	0.0076 (0.0071)	0.0283*** (0.0076)	0.0278** (0.0124)	0.0012 (0.0090)	0.0308*** (0.0095)
Treated × Many treated	0.0188* (0.0099)	0.0038 (0.0071)	0.0197*** (0.0074)	0.0089 (0.0123)	0.0003 (0.0092)	0.0166* (0.0092)
catcwadsother	X	X	X	X	X	X
catcwadsother	X	X	X	X	X	X
Strata × sub-experiment FE	X	X	X	X	X	X
Agency × sub-experiment FE	X	X	X	X	X	X
Caseworker FE	X	X	X	X	X	X
Observations	14,654	14,654	14,654	14,654	14,654	14,654
R2	0.07420	0.07321	0.06515	0.08036	0.08130	0.06790
Control mean	0.1054	0.0529	0.0541	0.1710	0.0883	0.0885
Mean concurrent ads (preselection)	4.150	4.150	4.150	4.150	4.150	4.150

Notes: OLS regressions of vacancy filling outcomes (indicator for at least one hire within 45 or 90 days) on fully interacted treatment status × tercile indicators of the treatment share among the caseworker's concurrent caseload. Tercile breaks are computed separately for the periods before and after the treatment share increase on February 6. The reference category is control vacancies in the lowest tercile (few treated concurrent vacancies). Concurrent caseload is measured as the number of other preselection-mode vacancies assigned to the same caseworker within a 7-day rolling window centred on the vacancy's creation date, excluding the vacancy itself. One observation is one job posting. Standard errors clustered at the establishment level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A8: Placebo Test: Pre-Treatment Period

		Vacancy filled					
		(1)	(2)	(3)	(4)	(5)	(6)
1	Horizon:	15 days	30 days	45 days	60 days	75 days	90 days
2							
3	Treated	0.0019 (0.0032)	0.0037 (0.0047)	-0.0003 (0.0058)	-0.0037 (0.0065)	-0.0005 (0.0068)	-0.0029 (0.0070)
5	Strata × sub-experiment FE	X	X	X	X	X	X
6	Agency × sub-experiment FE	X	X	X	X	X	X
7							
8	Observations	18,615	18,615	18,615	18,615	18,615	18,615
9	R2	0.01000	0.01333	0.01619	0.01602	0.01789	0.01889
10	Control mean	0.0387271089762834	0.0822575803062144	0.1262383668568	0.158060642449715	0.17472230561393	0.187481236865806

Notes: Placebo test using vacancies from November 2023 to April 2024, before the experiment began. Treatment status is assigned based on the establishment's eventual treatment assignment. Specifications include strata × sub-experiment and agency × sub-experiment fixed effects. Standard errors clustered at the establishment level in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A9: Treatment Effect on Number of Candidates Presented to Firm

	Number of candidates presented to firm (all)								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
1 Horizon:	3 days	5 days	10 days	15 days	30 days	45 days	60 days	75 days	90 days
2									
3 Treated	0.0514 (0.0353)	0.0461 (0.0329)	0.0445 (0.0293)	0.0436 (0.0277)	0.0466* (0.0269)	0.0398 (0.0282)	0.0328 (0.0304)	0.0466 (0.0317)	0.0299 (0.0338)
5 Strata x subexperiment FE	X	X	X	X	X	X	X	X	X
6 Unempl. office x subexperiment FE	X	X	X	X	X	X	X	X	X
7									
8 Observations	16,885	16,694	16,264	15,705	13,688	11,962	10,036	8,128	6,348
9 Control mean	2.35328618250819	3.10427528675704	4.427734375	5.29095220722364	6.5438843141193	7.10917127071823	7.3359489946052	7.37821398949406	7.56558441558442

Notes: PPML regression estimates. Outcome: number of candidates presented to the firm within the specified number of days after the vacancy enters preselection mode. One observation is one job posting. All specifications include strata $\times$ subexperiment and agency $\times$ subexperiment fixed effects. Standard errors clustered at the establishment level in parentheses. Subexperiment denotes an indicator variable for whether the preselection process for the vacancy started after February 6, 2025, when the treatment share was increased from 50% to 75%. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A10: Treatment effect by vacancy filling difficulty and caseworker workload composition (90-day horizon).

	Easy to fill			Hard to fill		
	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Var.:	All	Sourcing	Screening	All	Sourcing	Screening
Treated	0.0619*** (0.0217)	0.0270 (0.0165)	0.0384** (0.0175)	0.0436** (0.0182)	0.0263* (0.0137)	0.0223* (0.0131)
Concurrent ads same CW	-0.0026* (0.0013)	-8.68e-5 (0.0010)	-0.0025** (0.0011)	-0.0004 (0.0011)	-0.0001 (0.0009)	-0.0003 (0.0008)
Conc. ads with preselection flag same CW	-0.0011 (0.0023)	-0.0014 (0.0018)	-6.63e-5 (0.0017)	-0.0020 (0.0017)	-0.0013 (0.0013)	-0.0005 (0.0012)
Share treated among conc. ads with flag	0.0322 (0.0277)	-0.0047 (0.0205)	0.0449* (0.0230)	0.0323 (0.0232)	0.0256 (0.0178)	0.0059 (0.0165)
Treated $\times$ Share treated	-0.0698** (0.0339)	-0.0270 (0.0257)	-0.0395 (0.0273)	-0.0596** (0.0287)	-0.0417* (0.0222)	-0.0241 (0.0203)
Strata $\times$ sub-experiment FE	X	X	X	X	X	X
Agency $\times$ sub-experiment FE	X	X	X	X	X	X
Caseworker FE	X	X	X	X	X	X
Observations	6,847	6,847	6,847	6,564	6,564	6,564
R2	0.13557	0.12561	0.12497	0.12345	0.12910	0.10802
Control mean	0.2168	0.1071	0.1177	0.1297	0.0687	0.0651
Mean concurrent ads (preselection)	4.070	4.070	4.070	4.180	4.180	4.180

Notes: OLS regressions of vacancy filling outcomes (dummy for at least one hire within 90 days) on treatment status, share of treated ads among caseworker's concurrent workload, and their interaction. Concurrent workload is measured as the rolling sum of preselection-mode vacancies assigned to the same caseworker within a centered 7-day window around the vacancy's creation date, excluding the vacancy itself. "Easy to fill" and "Hard to fill" are defined by a median split on the IEE score (France Travail's prediction of the probability of filling the vacancy within 30 days). All specifications include strata $\times$ sub-experiment, agency $\times$ sub-experiment, and caseworker fixed effects. Standard errors clustered at the establishment level in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A11: Match quality of caseworker referrals: detailed indicators.

	All candidates			Sourcing			Screening		
	N	Control mean	Coef. (p)	N	Control mean	Coef. (p)	N	Control mean	Coef. (p)
<i>A. Occupation match</i>									
Job title match	135,323	0.144	0.002 (0.681)	38,441	0.221	0.002 (0.829)	96,882	0.115	0.000 (0.952)
<i>B. Municipality and distance</i>									
Municipality match	134,145	0.229	0.007 (0.261)	38,024	0.279	0.008 (0.397)	96,121	0.210	0.008 (0.247)
Distance (log km)	127,384	2.626	-0.030* (0.076)	36,426	2.366	-0.019 (0.306)	90,958	2.727	-0.030 (0.135)
<i>C. Experience</i>									
No exp. required × No experience	129,495	0.073	0.005* (0.087)	37,802	0.082	0.002 (0.725)	91,693	0.069	0.007** (0.035)
No exp. required × Has experience	129,495	0.364	0.011 (0.386)	37,802	0.420	0.011 (0.463)	91,693	0.342	0.011 (0.417)
Experience required × No experience	128,744	0.060	-0.001 (0.702)	37,870	0.045	-0.002 (0.481)	90,874	0.066	-0.000 (0.969)
Experience required × Has experience	128,744	0.456	-0.017 (0.204)	37,870	0.435	-0.009 (0.553)	90,874	0.465	-0.021 (0.138)
Experience duration meets requirement	122,916	0.858	0.006 (0.211)	37,231	0.888	0.006 (0.303)	85,685	0.846	0.006 (0.300)
<i>D. Formation</i>									
Education meets vacancy requirement	49,569	0.353	0.004 (0.548)	13,512	0.397	0.009 (0.453)	36,057	0.337	-0.000 (0.995)
Education missing (vacancy has requirement)	49,966	0.007	0.001 (0.542)	13,593	0.006	0.000 (0.858)	36,373	0.008	0.001 (0.446)
<i>E. Driving licence</i>									
No licence required × No licence	135,323	0.439	0.012 (0.267)	38,441	0.427	0.003 (0.804)	96,882	0.443	0.015 (0.193)
No licence required × Has licence	135,323	0.250	0.002 (0.729)	38,441	0.253	0.006 (0.421)	96,882	0.249	0.001 (0.937)
Licence desired × No licence	75,348	0.099	0.014* (0.080)	22,131	0.109	0.010 (0.307)	53,217	0.095	0.017* (0.066)
Licence desired × Has licence	101,085	0.058	-0.000 (0.934)	28,381	0.070	-0.005 (0.498)	72,704	0.053	0.001 (0.842)
Licence required × No licence	75,348	0.200	-0.017* (0.084)	22,131	0.171	-0.007 (0.474)	53,217	0.212	-0.019* (0.085)
Licence required × Holds required licence	120,161	0.104	-0.012* (0.090)	34,557	0.110	-0.008 (0.330)	85,604	0.101	-0.015* (0.064)
<i>F. Contract type</i>									
Wants CDI × Vacancy is CDI	127,737	0.503	-0.010 (0.495)	37,701	0.540	-0.022 (0.163)	90,036	0.488	-0.005 (0.748)
Wants CDD × Vacancy is CDD	131,635	0.053	0.005 (0.112)	38,044	0.059	0.006 (0.125)	93,591	0.051	0.004 (0.202)
Wants CDI × Vacancy is CDD	131,635	0.271	0.002 (0.896)	38,044	0.265	0.006 (0.686)	93,591	0.274	-0.000 (0.985)
<i>G. Working time</i>									
Working hours compatible	135,323	0.833	-0.006* (0.085)	38,441	0.879	0.010** (0.037)	96,882	0.815	-0.014*** (0.001)
<i>H. Salary</i>									
Job pays above reservation wage: log gap	22,310	0.182	-0.017* (0.060)	7,372	0.184	-0.030*** (0.004)	14,934	0.181	-0.013 (0.208)
Job pays below reservation wage: log gap	7,824	0.715	-0.097** (0.018)	2,411	0.693	-0.084 (0.180)	5,403	0.725	-0.112** (0.013)
Wage comparison missing	135,323	0.780	-0.010 (0.418)	38,441	0.737	0.009 (0.555)	96,882	0.797	-0.018 (0.155)

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: OLS regressions of match quality indicators on the treatment indicator, with strata × sub-experiment and agency × sub-experiment fixed effects. Each row is a separate regression. The unit of observation is a caseworker referral (mise en relation, MER). Column (1) pools all referrals. Column (2) restricts to caseworker-initiated referrals (sourcing channel). Column (3) restricts to job-seeker-initiated referrals approved by the caseworker (screening channel). Match quality indicators are constructed by linking each referral to the job seeker's online profile (DCO_PROFILPROFESSIONNEL_V) and the vacancy characteristics from the France Travail offer database. Appellation match equals one if the job seeker's desired ROME v3 appellation matches the appellation of the vacancy. Experience cross-indicators decompose referrals by the vacancy's experience requirement (none / desired / required) and whether the job seeker reports any relevant experience. Formation match is defined only for vacancies that specify an education requirement, and equals one if the job seeker's education level meets that requirement. Licence match indicators decompose referrals by the vacancy's driving licence requirement stringency and whether the job seeker holds the required licence. Contract and working-time indicators capture alignment between job seeker preferences and vacancy characteristics. The log wage difference is $\log(\text{ad max wage}) - \log(\text{reservation wage})$, both converted to annual full-time equivalent; positive values indicate the vacancy pays above the job seeker's stated reservation wage. Standard errors clustered at the establishment level in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A12: Balance across treatment status and caseworker tenure.

		Exp. CW, control	Coefficient (ref: experienced CW, control)		
	Model	Mean	Treated	New CW	Treated $\times$ New CW
Ad characteristics					
IEE score	OLS	0.465	0.008*	-0.010	-0.001
Min. salary (annual FTE)	OLS	23863.608	-70.722	-477.668	340.523
Max. salary (annual FTE)	OLS	29011.320	-1371.914**	474.126	-398.463
Posting firm size					
Size: 10 or less	OLS	0.380	0.005	0.035	-0.022
Size: 11–25	OLS	0.138	-0.012	-0.014	0.022
Size: 26–50	OLS	0.160	0.002	0.013	-0.030
Size: more than 50	OLS	0.244	-0.014	-0.038	0.042
Size: missing	OLS	0.077	0.019**	0.004	-0.013
Posting firm industry					
Sector: Construction	OLS	0.053	-0.002	0.007	-0.009
Sector: Manufacturing	OLS	0.101	0.017*	-0.035*	0.014
Sector: Commerce	OLS	0.143	-0.006	0.008	-0.013
Sector: Services	OLS	0.145	0.013	0.008	0.052
Sector: Health	OLS	0.194	-0.017	-0.004	-0.049
Pre-period hire rates					
Hires via portal per ad — all (45d)	OLS	0.089	0.017**	0.066*	-0.005
Hires via portal per ad — all (90d)	OLS	0.156	0.001	0.027	0.019
Hires via portal per ad — sourcing (45d)	OLS	0.049	0.011**	0.034	0.011
Hires via portal per ad — sourcing (90d)	OLS	0.094	-0.004	0.005	0.038
Hires via portal per ad — screening (45d)	OLS	0.044	0.008	0.037	-0.017
Hires via portal per ad — screening (90d)	OLS	0.072	0.006	0.022	-0.013
Pre-period counts					
Ads posted (pre-period)	Poisson	22.778	-0.211	0.166	0.201
Ads in preselection (pre-period)	Poisson	3.848	-0.011	-0.618**	0.178
Referrals per ad — all (45d)	Poisson	6.720	0.009	-0.145	0.043
Referrals per ad — sourcing (45d)	Poisson	1.929	0.027	-0.196*	0.095
Referrals per ad — screening (45d)	Poisson	4.792	0.002	-0.123	0.020
Total hires among unemployed	Poisson	10.968	-0.041	0.040	-0.004
Total hires — DPAE	Poisson	20.576	-0.066	-0.074	-0.222

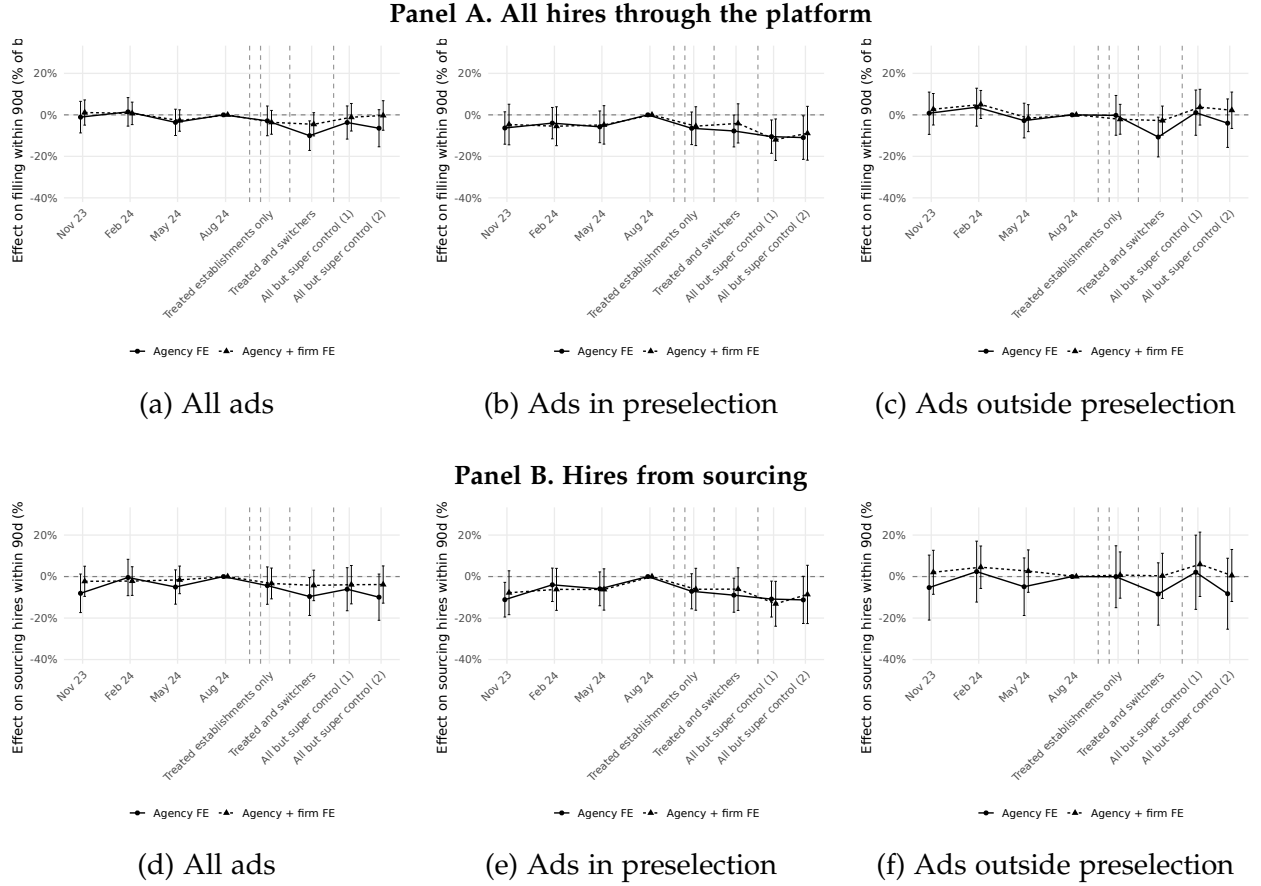
Notes: Each row reports the control mean for experienced caseworkers (tenure ≥ 1 year) alongside three regression coefficients from a single regression of the row variable on indicators for treated vacancy, new caseworker (tenure < 1 year), and their interaction, with strata $\times$ sub-experiment and agency $\times$ sub-experiment fixed effects. The reference category is a control vacancy handled by an experienced caseworker. Continuous and binary outcomes are estimated by OLS. Count outcomes (pre-period ad counts, referral counts, and aggregate hires) are estimated by Poisson quasi-maximum likelihood; In all regression, one experiment ad is one observation. Pre-period outcomes are computed over the twelve months prior to the first pilots of the experiment start date (September 2024). Referral counts and hire rates are averaged across all ads posted by the establishment in the pre-period; aggregate hire counts (DSN and DPAAE) are totals. Standard errors clustered at the establishment level. Stars denote significance at the *** 1%, ** 5%, * 10% level.

Table A13: Treatment effect controlling for caseworker value added

	Vacancy filled					
	(1)	(2)	(3)	(4)	(5)	(6)
Horizon:	15 days	30 days	45 days	60 days	75 days	90 days
<i>Panel A. Baseline</i>						
Treated	0.0043 (0.0033)	0.0128** (0.0051)	0.0149** (0.0062)	0.0145** (0.0068)	0.0152** (0.0072)	0.0104 (0.0076)
Strata $\times$ sub-experiment FE	X	X	X	X	X	X
Agency $\times$ sub-experiment FE	X	X	X	X	X	X
Observations	16,103	16,103	16,103	16,103	16,103	16,103
R2	0.01823	0.01965	0.02111	0.02544	0.02933	0.03009
Control mean	0.031	0.071	0.109	0.139	0.161	0.178
<i>Panel B. Controlling for value added</i>						
Treated	0.0045 (0.0033)	0.0127** (0.0051)	0.0145** (0.0062)	0.0139** (0.0068)	0.0146** (0.0072)	0.0100 (0.0075)
Caseworker value added (SD, same horizon)	0.0112*** (0.0026)	0.0191*** (0.0037)	0.0204*** (0.0043)	0.0254*** (0.0044)	0.0301*** (0.0047)	0.0334*** (0.0048)
Strata $\times$ sub-experiment FE	X	X	X	X	X	X
Agency $\times$ sub-experiment FE	X	X	X	X	X	X
Observations	16,103	16,103	16,103	16,103	16,103	16,103
R2	0.02047	0.02222	0.02328	0.02852	0.03322	0.03464
Control mean	0.031	0.071	0.109	0.139	0.161	0.178

Notes: OLS regressions of vacancy filling on treatment status, with strata $\times$ sub-experiment and agency $\times$ sub-experiment fixed effects, estimated on the experimental vacancies whose caseworker has a value added (93.5% of the sample). Panel A is the baseline specification on that sample; Panel B adds caseworker value added as a control. Caseworker value added is estimated on the vacancies of the year preceding the experiment (21 November 2023 to 20 November 2024) following equation (4), and is expressed in standard deviations of its distribution across caseworkers. It is constructed separately for each horizon, on the pre-period outcome measured over that same horizon, so that in every column the control is measured on the same outcome as the dependent variable; the normalising standard deviation therefore differs across columns. The unit of observation is one vacancy posting. Standard errors clustered at the establishment level in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure A4: Full roll-out: event study on job filling within 90 days



Notes: As Figure ??, with the outcome measured within 90 rather than 45 days of the ad being posted. Each panel plots the coefficients β_b of the event study described in Section ??: the difference in the filling probability between ads handled by experiment agencies and ads handled by the super control agencies of Normandy and Brittany, by time bin and relative to the last quarter before the first deployment. Columns differ by sample and Panel B restricts the outcome to hires that follow a caseworker referral obtained through sourcing. As there, each coefficient and confidence bound is divided by the baseline filling probability of its own panel, so that the vertical axis reads as a percentage of the pre-roll-out filling probability; those baselines are higher here than at 45 days, and the vertical range is shared with Figure ?. Solid lines report the specification with group, agency and difficulty $\times$ bin fixed effects; dashed lines add firm fixed effects. Because contracts are observed until October 31, 2025, ads posted in the last month of the sample cannot be observed over a full 90-day window and are excluded from this figure, which shortens the final bin relative to Figure ?. Error bars represent 95% confidence intervals from standard errors clustered at the establishment level.

B Survey results

B.1 Survey among caseworkers in experiment regions

Block 1: Preselection Practices (All Respondents)

Q1: Factors considered when deciding to offer preselection service [All resp.]

Nature of the vacancy (anticipated difficulty, specific prerequisites) ($N = 77$): Predominant role: 79% / Can influence: 19% / Not considered: 1%

Employer characteristics (client status, HR department presence) ($N = 77$): Predominant role: 36% / Can influence: 47% / Not considered: 17%

Agency objectives ($N = 77$): Predominant role: 22% / Can influence: 49% / Not considered: 29%

Agency capacity ($N = 77$): Predominant role: 25% / Can influence: 42% / Not considered: 34%

Q2: Types of vacancies for which you propose preselection [All resp.]

Vacancy with many expected applications ($N = 77$): Always propose: 60% / May propose: 34% / Never propose: 6%

Vacancy anticipated to be hard to fill ($N = 77$): Always propose: 22% / May propose: 58% / Never propose: 19%

Vacancy with specific prerequisites ($N = 77$): Always propose: 66% / May propose: 32% / Never propose: 1%

Employer has no HR department ($N = 77$): Always propose: 35% / May propose: 64% / Never propose: 1%

Employer is a regular client ($N = 77$): Always propose: 19% / May propose: 74% / Never propose: 6%

Other ($N = 77$): Always propose: 31% / May propose: 26% / Never propose: 43%

Q3: Objectives of preselection [All resp.]

Filter candidates on attractive vacancies ($N = 77$): Predominant objective: 49% / Can be an objective: 44% / Not an objective: 6%

Search candidates for hard-to-fill vacancies ($N = 77$): Predominant objective: 44% / Can be an objective: 44% / Not an objective: 12%

Avoid irrelevant applications ($N = 77$): Predominant objective: 88% / Can be an objective: 12% / Not an objective: 0%

Improve employer relationship ($N = 77$): Predominant objective: 57% / Can be an ob-

jective: 42% / Not an objective: 1%

Other ($N = 77$): Predominant objective: 27% / Can be an objective: 22% / Not an objective: 51%

Q4 [All resp.] Propose preselection if vacancy is struggling after several days. $N = 77$

Yes, always: 6% · Yes, often: 32% · Rarely: 14% · Not based on this: 47%

Block 2: MatchFT Adoption

Q5 [All resp.] Have you used MatchFT?. $N = 86$

Yes: 88% · No: 12%

Q6: Reasons for not using MatchFT [Non-users only]

Don't see the point ($N = 8$): Yes: 25%

Don't want to contact job seekers this way ($N = 8$): Yes: 38%

Don't use the SDR tool ($N = 8$): Yes: 50%

Don't know what MatchFT is ($N = 8$): Yes: 50%

Other ($N = 8$): Yes: 38%

Block 3: MatchFT Usage (Users Only)

Q7: Types of vacancies where MatchFT is relevant [Users only]

All vacancies ($N = 72$): Yes: 21%

Vacancies with few/irrelevant spontaneous applications ($N = 72$): Yes: 93%

Vacancies with limited SDR results ($N = 72$): Yes: 76%

Vacancies with many candidates ($N = 72$): Yes: 22%

Don't find MatchFT relevant ($N = 72$): Yes: 12%

Other ($N = 72$): Yes: 31%

Q8: Positive aspects of MatchFT [Users only]

Contact via SMS ($N = 72$): Yes: 93%

Faster candidate response time ($N = 72$): Yes: 74%

Additional info from conversation details ($N = 72$): Yes: 68%

Ability to contact more people at once ($N = 71$): Yes: 87%

Higher response rate than other channels ($N = 71$): Yes: 30%

Other ($N = 71$): Yes: 28%

Q9: Actions to identify candidate profiles [Users only]

Check SDR score ($N = 72$): Always: 46% / Often: 36% / Rarely: 14% / Never: 4%

Consult dossier in MAP ($N = 72$): Always: 40% / Often: 49% / Rarely: 8% / Never: 3%

Discuss with caseworker in charge of job seeker ($N = 72$): Always: 3% / Often: 25% / Rarely: 60% / Never: 12%

Check CV ($N = 72$): Always: 69% / Often: 22% / Rarely: 4% / Never: 4%

Check profile summary from SDR ($N = 72$): Always: 62% / Often: 28% / Rarely: 7% / Never: 3%

Other ($N = 71$): Always: 15% / Often: 15% / Rarely: 0% / Never: 69%

Q10: Practice changes with MatchFT [Users only]

Propose vacancy to more candidates ($N = 72$): Yes: 57%

Propose vacancy to different candidates ($N = 72$): Yes: 54%

Other ($N = 71$): Yes: 24%

Q11: How MatchFT enables reaching different candidates [Users with Q10.2=Yes]

Candidates with lower SDR score ($N = 39$): Yes: 74%

Geographically more distant candidates ($N = 39$): Yes: 82%

Candidates with sparser dossiers ($N = 39$): Yes: 87%

Other ($N = 39$): Yes: 28%

Q12 [Users only] Delay before checking MatchFT responses. $N = 71$

Within 6h: 39% · Within 12h: 11% · Within 24h: 39% · Within 48h: 4% · Within a week: 6%

Q13 [Users only] Which candidacies do you review after prequalification. $N = 72$

All prerequisites validated: 18% · Partially/fully validated: 18% · Any exchange occurred: 64%

Q14 [Users only] How often do you read MatchFT conversations. $N = 72$

Always: 67% · Often: 22% · Sometimes: 7% · Never: 4%

Q15 [Users only] Time needed to verify prequalification elements per candidate. $N = 71$

<1 min: 21% · 1-5 min: 56% · >5 min: 23%

Q16.1 [Users only] Out of 10 candidates responding to prequalification, how many do you call. $N = 64$

Mean: 3.39, Median: 3, SD: 2.81

Q17: Reasons for calling candidates [Users who call (Q16>0)]

Confirm received information ($N = 56$): Yes: 80%

Provide information about the vacancy ($N = 56$): Yes: 84%
Obtain more information about candidate profile ($N = 56$): Yes: 91%
Other ($N = 56$): Yes: 38%

Q18 [Users who call ($Q16 > 0$)] Average duration of calls. $N = 56$

<1 min: 4% · 1-5 min: 70% · 5-10 min: 21% · >10 min: 5%

Q19 [Users only] Does MatchFT lead you to call more candidates. $N = 64$

Yes: 14% · No: 55% · Don't know: 31%

Q20 [Users with $Q19 = \text{Yes}$] Reasons for calling more candidates (open text). $N = 9$

[Open-ended responses not shown]

Q21 [Users only] Action if candidate does not respond to MatchFT. $N = 71$

Nothing: 62% · Send email: 32% · Try calling: 6%

Q22: Difficulties encountered with MatchFT [Users only]

Not enough candidate responses ($N = 72$): Yes: 86%

Technical problems ($N = 72$): Yes: 25%

Other ($N = 71$): Yes: 32%

Q23: Perceived improvement with MatchFT [Users only]

Quality of service for job seekers ($N = 72$): Improved: 49% / Unchanged: 47% / Worsened: 4%

Quality of service for employers ($N = 72$): Improved: 38% / Unchanged: 61% / Worsened: 1%

Time savings ($N = 72$): Improved: 56% / Unchanged: 35% / Worsened: 10%

Q24 [Users only] Remarks or suggestions for improving MatchFT (open text). $N = 70$

[Open-ended responses not shown]

Q25.1 [Users only] Overall satisfaction with MatchFT (0-5 scale). $N = 71$

Mean: 2.97, Median: 3, SD: 1.42

B.2 Survey among caseworkers in non-experiment regions

Block 1: Preselection Practices (All Respondents)

Q1: Factors considered when deciding to offer preselection service [All resp.]

Nature of the vacancy (anticipated difficulty, specific prerequisites) ($N = 91$): Predominant role: 86% / Can influence: 12% / Not considered: 2%

Employer characteristics (client status, HR department presence) ($N = 91$): Predominant

role: 53% / Can influence: 32% / Not considered: 15%

Agency objectives ($N = 91$): Predominant role: 24% / Can influence: 44% / Not considered: 32%

Agency capacity ($N = 91$): Predominant role: 24% / Can influence: 43% / Not considered: 33%

Q2: Types of vacancies for which you propose preselection [All resp.]

Vacancy with many expected applications ($N = 91$): Always propose: 55% / May propose: 43% / Never propose: 2%

Vacancy anticipated to be hard to fill ($N = 91$): Always propose: 21% / May propose: 63% / Never propose: 16%

Vacancy with specific prerequisites ($N = 91$): Always propose: 78% / May propose: 20% / Never propose: 2%

Employer has no HR department ($N = 91$): Always propose: 47% / May propose: 53% / Never propose: 0%

Employer is a regular client ($N = 91$): Always propose: 27% / May propose: 65% / Never propose: 8%

Other ($N = 91$): Always propose: 31% / May propose: 19% / Never propose: 51%

Q3: Objectives of preselection [All resp.]

Filter candidates on attractive vacancies ($N = 91$): Predominant objective: 48% / Can be an objective: 40% / Not an objective: 12%

Search candidates for hard-to-fill vacancies ($N = 91$): Predominant objective: 40% / Can be an objective: 40% / Not an objective: 21%

Avoid irrelevant applications ($N = 91$): Predominant objective: 85% / Can be an objective: 14% / Not an objective: 1%

Improve employer relationship ($N = 91$): Predominant objective: 68% / Can be an objective: 31% / Not an objective: 1%

Other ($N = 91$): Predominant objective: 36% / Can be an objective: 14% / Not an objective: 49%

Q4 [All resp.] Propose preselection if vacancy is struggling after several days. $N = 91$

Yes, always: 13% · Yes, often: 32% · Rarely: 11% · Not based on this: 44%

Block 2: Candidate Search Workflow

Q1 [All resp.] First action after receiving a vacancy in preselection mode. $N = 91$

Immediate search: 80% · Wait for applications: 7% · Other: 13%

Q5: Actions to identify candidate profiles [All resp.]

Check SDR score ($N = 91$): Always: 58% / Often: 29% / Rarely: 12% / Never: 1%

Consult dossier in MAP ($N = 91$): Always: 57% / Often: 35% / Rarely: 7% / Never: 1%

Discuss with caseworker in charge of job seeker ($N = 91$): Always: 7% / Often: 33% / Rarely: 55% / Never: 5%

Check CV ($N = 91$): Always: 82% / Often: 16% / Rarely: 0% / Never: 1%

Check profile summary from SDR ($N = 91$): Always: 68% / Often: 21% / Rarely: 10% / Never: 1%

Other ($N = 91$): Always: 26% / Often: 19% / Rarely: 5% / Never: 49%

Q6: Frequency of dossier/SDR consultation [All resp.]

Consult dossier in detail ($N = 91$): Always: 22% / Often: 57% / Rarely: 19% / Never: 2%

Check SDR systematically ($N = 91$): Always: 35% / Often: 40% / Rarely: 18% / Never: 8%

Q7 [All resp.] Targeting strategy when contacting candidates. $N = 91$

Target narrowly: 78% · Contact broadly: 22%

Q8 [All resp.] Number of candidates typically contacted per vacancy. $N = 89$

<10: 44% · 10-20: 46% · >20: 10%

Q9: Frequency of verification actions [All resp.]

Verify prerequisites ($N = 89$): Always: 31% / Often: 48% / Rarely: 19% / Never: 1%

Check availability ($N = 89$): Always: 53% / Often: 35% / Rarely: 8% / Never: 4%

Check motivation ($N = 89$): Always: 11% / Often: 31% / Rarely: 44% / Never: 13%

Check mobility ($N = 89$): Always: 1% / Often: 22% / Rarely: 43% / Never: 34%

Other ($N = 89$): Always: 11% / Often: 21% / Rarely: 9% / Never: 58%

Q10.1 [All resp.] Out of 10 candidates responding, how many do you contact further. $N = 88$

Mean: 3.48, Median: 3, SD: 1.85

Q11 [All resp.] Time spent per candidate for initial review. $N = 88$

1-5 min: 59% · 5-10 min: 36% · >10 min: 5%

Q12: Information verified when reviewing candidates [All resp.]

Confirm received information ($N = 88$): Yes: 100%
Provide information about vacancy ($N = 88$): Yes: 99%
Get more information about profile ($N = 88$): Yes: 100%
Other ($N = 88$): Yes: 62%

Q13 [All resp.] Delay before checking candidate responses. $N = 85$

Within 6h: 16% · Within 12h: 12% · Within 24h: 38% · Within 48h: 15% · Within a week: 19%

Q14: Level of detail when reviewing candidate elements [All resp.]

Review CV ($N = 85$): In detail: 41% / Quickly: 39% / Don't review: 20%

Review dossier ($N = 85$): In detail: 93% / Quickly: 5% / Don't review: 2%

Review motivation ($N = 85$): In detail: 27% / Quickly: 16% / Don't review: 56%

Q15 [All resp.] Time needed to verify information per candidate. $N = 89$

<1 min: 7% · 1-5 min: 72% · >5 min: 21%

Q16.1 [All resp.] Out of candidates reviewed, how many do you call. $N = 85$

Mean: 3.47, Median: 3, SD: 2.72

Q17: Reasons for calling candidates [Resp. who call ($Q16 > 0$)]

Confirm received information ($N = 78$): Yes: 85%

Provide information about the vacancy ($N = 78$): Yes: 71%

Obtain more information about candidate profile ($N = 78$): Yes: 94%

Verify availability ($N = 78$): Yes: 95%

Other ($N = 78$): Yes: 41%

Q18 [Resp. who call ($Q16 > 0$)] Average duration of calls. $N = 78$

<1 min: 3% · 1-5 min: 62% · 5-10 min: 29% · >10 min: 6%

Q19: Decision criteria for candidate selection [All resp.]

Based on CV match ($N = 84$): Yes: 89%

Based on availability ($N = 84$): Yes: 69%

Other ($N = 84$): Yes: 46%

Q20 [All resp.] Primary contact channel used. $N = 91$

Work landline: 9% · Work mobile: 34% · GMS/GEM by email: 11% · GMS/GEM by SMS: 12% · Job offer by email: 26% · Other: 8%

Q21 [All resp.] Self-reported effectiveness score (0-20 scale). $N = 91$

Mean: 9.46, Median: 8, SD: 5.41